APPLICATION OF RANDOM FOREST MODEL FOR MILK QUALITY CLASSIFICATION AND PROTOTYPE SYSTEM DEVELOPMENT IN UGANDAN DAIRY PROCESSING PLANTS

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Declaration

I, Behangana Emmanuel, hereby declare that this dissertation titled "Application of Random Forest Model for Milk Quality Classification and Prototype System Development in Ugandan Dairy Processing Plants" is my original work and has not been submitted for any degree award in any other university or institution of higher learning.

All sources of information and materials used in this research have been duly acknowledged through proper citations and references.

Where collaborative work or assistance was received, it has been explicitly stated and acknowledged in the appropriate sections of this thesis.

____________________ 16/02/2025

Behangana Emmanuel DATE

Approval

I hereby certify that this research thesis entitled "Application of Random Forest Model for Milk Quality Classification and Prototype System Development in Ugandan Dairy Processing Plants" has been prepared and submitted by Behangana Emmanuel under my supervision. I approve it for submission to the **Faculty of Agriculture, Environment Science, and Technology**, Bishop Stuart University, in fulfillment of the requirements for the award of Master of Business in Information Technology.

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Dedication

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Abstract

Milk quality remains a major concern in Ugandan dairy processing plants due to post-harvest losses, inconsistent handling practices, and the continued reliance on manual quality assessment methods that are often slow, subjective, and insufficient for timely decision-making. These challenges are particularly significant in key dairy-producing districts such as Mbarara and Kiruhura, where maintaining milk quality is essential for food safety, market competitiveness, and public health. This study was undertaken to develop and evaluate a Random Forest model for milk quality classification and to integrate the model into a prototype system to support decision-making in Ugandan dairy processing plants.

The study used historical milk quality records comprising 1,059 observations and seven key physicochemical and sensory attributes, namely pH, temperature, taste, odor, fat content, titratable acidity, protein content, lactose content, Total plate count, somatic cell count, and color. Data preprocessing was undertaken to prepare the dataset for machine learning classification. Several classification models were evaluated during experimentation, including Support Vector Machine, Artificial Neural Network, XGBoost, and Random Forest. Among the evaluated models, Random Forest achieved the best overall classification performance and was therefore selected as the principal model for the study. The selected model was then integrated into a prototype desktop-based interface developed to classify milk quality into low, medium, and high categories.

The findings demonstrate that the Random Forest model provides a practical approach for supporting milk quality classification in dairy processing environments where rapid and consistent interpretation of quality indicators is required. The prototype system offers an interactive interface through which plant operators can enter quality-related parameters and obtain classification outputs that may support routine quality assessment. The study contributes to the application of machine learning in agricultural value chains and provides a foundation for the development of practical decision-support tools for the Ugandan dairy sector.

Keywords: Random Forest, milk quality classification, prototype system development, dairy processing plants, Uganda, machine learning.

List of Acronyms

ML	Machine Learning
AI	Artificial Intelligence
GS	Google Scholar
Py	Python Programming
Qt	Quick Designer Tool
SVM	Support Vector Machine
ANN	Artificial Neural Networks
PCA	Principal Component Analysis
RF	Random Forest
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
KNN	K-Nearest Neighbors
DW-KNN	Distance -Weighted KNN
GBM	Gradient Boosting Machine
E-Nose	Electronic Nose
LDA	Linear Discriminant Analysis
AdaBoost	Adaptive Boosting
MCCs	Milk Collection Centers
DDA	Dairy Development Authority

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Chapter One

Introduction

1.1 Introduction

The dairy sector plays a significant role in Uganda's economy through its contribution to household income, food security, employment, and agro-industrial development (Ekou, 2014; FAO, 2011). Milk and milk products are important to nutrition and livelihoods, especially in the cattle corridor districts where dairy production is a major economic activity. However, the sector continues to face challenges related to milk quality deterioration during handling, transportation, collection, and processing. These challenges reduce the value of milk, expose consumers to health risks, and limit the competitiveness of dairy enterprises in domestic and export markets (Grimaud et al., 2007; Nyokabi et al., 2021).

In response to such challenges, machine learning has emerged as a useful data-driven approach for solving prediction and decision-support problems. In supervised machine learning, algorithms learn patterns from historical data containing input features and known output labels and then use those learned patterns to predict the class of new observations (Shahzad et al., 2025). Common supervised machine learning techniques include decision trees, support vector machines, artificial neural networks, and ensemble methods such as Random Forest. These techniques are particularly useful where large amounts of structured data are available and where human judgment alone may produce inconsistent outcomes.

Within machine learning, this study was concerned with a classification problem. A classification problem raised when the aim is to assign observations to predefined discrete categories rather than to predict continuous numerical values. In the context of milk quality assessment, the task was to use measured milk characteristics such as pH, temperature, taste, odor, fat content, titratable acidity, protein content, lactose content, Total plate count, somatic cell count, and color to classify milk into quality classes such as low, medium, or high quality. This made the problem suitable for supervised classification techniques because the output variable consists of categorical class labels.

Although traditional milk quality assessment methods are still used in many Ugandan dairy processing contexts, they are often time-consuming, operator-dependent, and less suitable for consistent decision-making in increasingly complex supply chains (Ahikiriza et al., 2021). This created the need not only for an accurate classification model but also for a practical system through which dairy personnel can interact with the model and use its outputs in routine operations.

For that reason, the study extended beyond model development to the design of a system prototype. A prototype is an early representation of an intended software system that is used to demonstrate functionality, refine user requirements, and guide eventual implementation. It ranged from a low-fidelity conceptual design to a high-fidelity interactive interface that resembles the final system. In this study, the prototype system served as a user-oriented decision-support tool through which milk quality data was entered, classification results were displayed, and the practical applicability of the model was assessed in the dairy processing environment.

This study applied supervised machine learning to the problem of milk quality classification in Ugandan dairy processing plants, with particular focus on the Random Forest model. Random Forest is emphasized because of its suitability for structured tabular data, its robustness, and its usefulness in classification tasks. The study further integrated the selected model into a prototype system so as to demonstrate how machine learning outputs can be translated into a practical decision-support tool for dairy quality assessment. By combining algorithmic classification with prototype system development, the study addresses both classification performance and practical usability in the Ugandan dairy processing context.

1.2 Background to the Study

Uganda is one of the leading milk-producing countries in East Africa, and the dairy industry has expanded steadily over the years (Ssekibaala, 2019). Major dairy-producing districts such as Mbarara and Kiruhura contribute substantially to national milk output and serve as important centers for milk collection, transportation, and processing. Despite this growth, maintaining milk quality remains a persistent challenge due to inadequate cold-chain management, delays in transportation, poor handling practices, contamination risks, and variability in testing procedures (Kilima et al., 2024). These problems contribute to quality losses along the dairy value chain and affect processors, traders, farmers, and consumers alike.

Milk quality is commonly determined using a combination of physicochemical and sensory indicators. In this study, the variables considered include pH, temperature, taste, odor, fat content, titratable acidity, protein content, lactose content, Total plate count, somatic cell count, and color. These variables are important because they capture characteristics associated with freshness, contamination, spoilage, adulteration, and market acceptability (Moradi et al., 2023). In practical settings, however, interpreting such indicators consistently can be difficult when assessment depends primarily on manual judgment. This creates a need for analytical approaches capable of learning patterns from existing quality records and generating consistent classification outcomes.

Globally, machine learning has increasingly been applied in agriculture and food quality management to support classification, quality monitoring, and decision-making (Shine & Murphy, 2021). In the context of dairy quality assessment, machine learning models can analyze multiple quality attributes simultaneously and identify relationships that may not be easily detected through manual inspection alone. Within the present study, different machine learning models were evaluated, and Random Forest emerged as the best-performing model among the tested alternatives. Its selection for this study is therefore grounded in empirical performance within the dataset used, as well as its suitability for classification tasks involving structured tabular data.

Beyond algorithmic performance, there was also a practical need to translate model outputs into a usable operational tool. A machine learning model has limited value in practice if it is not presented through an interface that supports end-user interaction. For this reason, the study included prototype system development as an integral component. The prototype intended to provide an interactive means through which users can input milk quality variables results from the laboratory and obtain classification outputs to support routine decision-making in dairy processing plants.

1.3 Problem Statement

Milk quality management remains a major challenge in Ugandan dairy processing plants despite the sector's importance to national development and household livelihoods (Sugino et al., 2023). (Opportunities for Innovation and Intervention in Uganda’s Dairy Value Chain, 2023) study highlights that even when farm-level practices improve, the broader value chain, including processing and collection centers,

fails to meet hygiene standards, with only 13.3% of samples at collection centers meeting criteria for human consumption (Sugino et al., 2023). Furthermore, the Ugandan dairy sector contributes roughly 4.2% to the national GDP and 17% to the agricultural GDP, underscoring its role in national development and household livelihoods.

In many operational settings, milk quality assessment still relies on conventional procedures that may be slow, subjective, and insufficiently consistent for effective classification and decision-making. As a result, poor-quality milk may enter processing channels, while acceptable milk may be rejected or downgraded due to inconsistent assessment practices. These weaknesses contribute to economic losses, undermine consumer confidence, and create food safety concerns within the dairy value chain (Md. Shakhor Uddin et al., 2025).

Although machine learning methods have shown immense potential in food quality analysis, there remains a prominent practical gap in the application of context-appropriate classification models and user-oriented prototype systems for milk quality assessment in Ugandan dairy processing environments (Martelli, 2025). Currently, regional milk collection centers and processing nodes in Uganda still heavily rely on manual, rudimentary testing mechanisms like lactometers and alcohol stability tests, leaving a severe deficiency in digital, user-friendly classification frameworks tailored for local operators (Ariong, 2023; Ndambi, 2020). In particular, there was limited evidence of simple, usable machine learning-based tools designed to support classification of milk quality using locally relevant quality indicators. This gap was especially important in major dairy-producing areas such as Mbarara and Kiruhura, where routine and reliable quality assessment is essential for operational efficiency and product safety.

Accordingly, this study addressed the problem by developing and evaluating a Random Forest model for milk quality classification and by integrating the selected model into a prototype system intended to support milk quality decision-making in Ugandan dairy processing plants.

1.4 General Objective

The general objective of this study was to develop and evaluate a Random Forest model for milk quality classification and to integrate it into a prototype system for use in Ugandan dairy processing plants.

1.5 Specific Objectives

1. To identify the key milk quality features required for classification of milk into defined quality categories.
2. To develop and evaluate a Random Forest model for milk quality classification using appropriate classification metrics.

3. To design and implement a prototype system that applies the Random Forest model to support milk quality classification in Ugandan dairy processing plants.
4. To assess the usability and practical applicability of the prototype system in supporting milk quality assessment.

1.6 Research Questions

1. What key milk quality features should be identified and preprocessed for effective milk quality classification?
2. How effectively can the Random Forest model classify milk quality into the defined categories?
3. How can a prototype system be designed and implemented to apply the Random Forest model for milk quality classification in dairy processing plants?
4. How usable and practically applicable is the prototype system for supporting milk quality assessment by plant operators?

1.7 Significance of the Study

This study is significant to dairy processors because it provides a data-driven approach for improving consistency in milk quality classification. By supporting more systematic interpretation of milk quality indicators, the study has the potential to reduce uncertainty in assessment decisions and improve quality control practices within dairy processing operations.

The study is also significant to farmers, consumers, and regulatory stakeholders. Improved classification of milk quality may contribute to better handling of milk entering the value chain, enhanced food safety, and greater confidence in dairy products. In broader terms, the study may support efforts aimed at improving efficiency, product quality, and competitiveness in Uganda's dairy sector.

From an academic perspective, the study contributes to the growing body of knowledge on the application of machine learning in agricultural and food systems within developing-country contexts. It further demonstrates how machine learning classification can be linked to prototype system development to produce practically relevant decision-support tools rather than purely theoretical models.

1.8 Scope of the Study

1.8.1 Content Scope

The study focused on milk quality classification using selected milk quality indicators, namely pH, temperature, taste, odor, fat content, titratable acidity, protein content, lactose content, Total plate count, somatic cell count, and color. It covered data preprocessing, development and evaluation of the Random Forest classification model, and the design and implementation of a prototype desktop-based system for user interaction with the model. The study was limited to classification of milk into defined quality categories and did not extend to sensor automation, live-streaming data integration, or industrial-scale deployment.

1.8.2 Geographical Scope

The study was conducted in the Ugandan dairy context, with focus on dairy processing plants and related milk quality assessment settings in Mbarara and Kiruhura districts, which are among the major milk-producing areas in the country.

1.8.3 Time Scope

The study was confined to the period during which the data were obtained, processed, modeled, and translated into a prototype system for academic research purposes. The work specifically addressed the use of historical milk quality records available for the study and their application in model development and prototype design.

Chapter Two

Literature Review

2.1 Introduction

This chapter presents a comprehensive review of existing literature related to the prediction of milk quality using machine learning (ML) models, particularly within the context of dairy processing. The review is organized around the main themes derived from the study's objectives: data collection and feature selection, machine learning model development and evaluation, and validation with user interface integration.

To ensure conceptual clarity and provide a firm foundation for the review, the following key terms were used in the study as defined:

Milk Quality: This refers to the compositional, microbiological, and sensory characteristics of milk that determine its safety, acceptability, and suitability for processing and consumption. Quality attributes include pH, fat content, protein levels, bacterial counts, somatic cell count, and sensory parameters such as taste, odor, and color.

Machine Learning (ML): A subset of artificial intelligence (AI) refers to computer algorithms that learn from data to identify patterns and make predictions or decisions without being explicitly programmed. In the context of this study, ML was used to analyze datasets from EA standards 2019 raw milk quality parameters to predict milk quality classifications.

Milk Processing Plants: These are industrial facilities where raw milk is collected, tested, pasteurized, processed into various dairy products, and packaged for distribution. In Uganda, these plants range from small-scale local operations to large commercial processors.

Feature Selection: This is the process of identifying the most relevant input variables (features) from a dataset that contribute significantly to the predictive output of an ML model. Effective feature selection enhances model accuracy and efficiency.

Predictive Analytics: This is the use of data, statistical algorithms, and machine learning techniques to identify the likelihood of future outcomes based on historical data. In this study, predictive analytics is applied to forecast milk quality before physical testing

Data Preprocessing: This includes all the necessary steps taken to clean, normalize, and transform raw data into a suitable format for machine learning models. It ensures that the input data is accurate, complete, and structured appropriately for analysis

Real-Time Monitoring: A process by which systems continuously collect and analyze data, allowing immediate feedback and action. Applied to milk quality, this enables early detection of deviations from quality standards during processing.

Ugandan Context: Refers to the specific operational, economic, environmental, and infrastructural conditions found within Uganda's dairy industry. These include issues like inconsistent electricity supply, manual data collection, limited digital infrastructure, and variability in farmer and staff training levels.

2.2 Data Collection and Feature Selection

A Kaggle dataset of 1,059 entries and eight features pH, turbidity, temperature, fat rate, taste, odor, and grade was used by (Samad & Taze, 2024) [15]. Milk quality was divided into three categories by the grading system: low, medium, and high. Relevant characteristics for milk quality were found by feature selection. To increase classification accuracy, machine learning models were developed and tested using all of these features.

To produce a comprehensive dataset, [5] gathered information from a variety of sources. Features including pH, temperature, taste, odor, fat, turbidity, and color are all included in the dataset. For the purpose of machine learning analysis, these characteristics were assessed and documented. The most pertinent characteristics for milk quality were found through feature selection. To increase classification accuracy, machine learning models were constructed and tested using all gathered characteristics. 1,000 raw milk samples were gathered from ten cattle farms in the province of Hebei by [14]. They used an E-nose system and DHI laboratory procedures to examine these materials. Milk fat, protein, lactose, total solids, somatic cells (SCC), and urea nitrogen were all assessed by the DHI analysis. The E-nose system used seven metal oxide semiconductor sensors to monitor volatile chemicals. They kept important characteristics from the DHI and E-nose data by using PCA and LDA for feature selection. Machine learning models for identifying milk sources and estimating their quality were created using these characteristics. The Kaggle Milk Quality dataset, comprising 1,059 samples with seven manually observed attributes pH, temperature, taste, odor, fat, turbidity, and color was utilized to classify milk quality into low, medium, and high categories; in this context, [19] conducted feature selection to identify the most elevant attributes and employed all seven features in training and testing various machine learning models to enhance classification accuracy.

The Kaggle milk classification database, which includes characteristics such as pH, turbidity, temperature, fat content, taste, and odor, was used to develop a dataset for machine learning analysis; in their study, [9] applied feature selection to determine the most relevant attributes for milk quality and utilized all collected features to train and evaluate machine learning models with the goal of enhancing classification accuracy

2.3 Model Development and Evaluation

For the purpose of detecting the quality of milk, [15] created two machine learning models: Distance-Weighted KNN (DW-KNN) and k-Nearest Neighbors (KNN). KNN categorizes samples according to the most common category among closest neighbors, however DW-KNN gives closer neighbors greater weight, increasing prediction accuracy. The models were assessed using F1 score, recall, accuracy, and precision. Distance weighting improves classification performance, as demonstrated by DW-KNN's superior performance over KNN.

Based on characteristics like pH, temperature, taste, flavor, fat, turbidity, and color, [5] employed machine learning methods, such as Random Forest and Support Vector Machine (SVM), to forecast milk quality. 80% of the data was used for training, and 20% was used for testing. To improve the models, hyperparameter tuning was used. F1 score, accuracy, precision, and recall were used to assess performance. The models were shown to be successful in classifying milk quality and improving evaluation reliability in the dairy business, with Random Forest achieving the best accuracy and SVM coming in second.

Using machine learning approaches, [14] created models for identifying milk sources and estimating milk quality. They employed the random forest (RF), logistic regression (LR), and support vector machine (SVM) techniques to identify the milk source. Using features from Dairy Herd Improvement analytical data, E-nose readings, and a mix of both, they used Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) to reduce dimensionality. The maximum accuracy was attained by the SVM model that used fusion features after LDA.

For milk quality estimation, [14] focused on predicting milk fat and protein content. The researchers used Gradient Boosting Decision Tree, Extreme Gradient Boosting, and Random Forest (RF) algorithms. The models were developed using E-nose measurements, and the RF model provided the best performance for milk fat and milk protein. The evaluation metrics included Mean Absolute Error (MAE), Mean Squared Error (MSE), and Coefficient of Determination (R^2), with the RF model showing the smallest estimation error and highest R^2 values, indicating its effectiveness in estimating milk quality parameters.

Neural Network (NN) and Adaptive Boosting (AdaBoost) algorithms were implemented using the Orange Data Mining Tool for preprocessing, training, testing, and visualization; in this analysis, [19] trained the models with 100 samples per class, where AdaBoost enhanced performance by combining weak classifiers into a strong one, and NN mimicked human brain learning processes. The models were evaluated based on classification accuracy, AUC, F1 score, precision, and recall, with AdaBoost demonstrating superior performance, greater stability, and faster convergence. Results, presented through ROC curves and metric values, highlighted AdaBoost's effectiveness in accurately classifying milk quality.

To classify milk quality based on attributes such as pH, turbidity, temperature, fat content, taste, and odor, three machine learning algorithms Random Forest, Extra Trees Classifier, and KNeighbors Classifier, were employed; in this study, [9] split the dataset into 80% for training and 20% for testing and applied GridSearchCV with 10-fold cross-validation for parameter optimization. Model performance was assessed using confusion matrices, focusing on accuracy, sensitivity, and precision. All three models achieved high accuracy with minimal differences in sensitivity and precision and demonstrated robust performance even with up to 50% noise added to the data.

2.3.1 Model Validation and User Interface Development

For model validation, [5] noted that it used a dataset available in the Kaggle repository. [14] Validated the developed models by using a five-fold cross-validation method to prevent over fitting. This method involves randomly dividing the training set into five subsets, using different subsets as the validation set each time to obtain the accuracy rate, and finally averaging the accuracy rates of each subset. The classification performance was evaluated based on the accuracy of the models on the testing set. The results showed that the SVM model with fusion features after LDA dimensionality reduction achieved the highest accuracy for milk source identification. For milk quality estimation, the RF model provided the best performance for milk fat and milk protein, indicating its effectiveness in estimating milk quality parameters.

2.3.2 Support Vector Machines (SVM) in Milk Quality Classification

Support Vector Machines (SVM) are supervised machine learning algorithms that are widely used for classification and regression tasks. Introduced by Vapnik in the 1990s, SVM operates by identifying the hyperplane that best separates data points into their respective categories while maximizing the margin between them. The support vectors are the data points closest to the hyperplane and are critical in defining its position and orientation. The SVM's ability to use kernel functions (e.g., radial basis function, polynomial kernel) allows it to handle non-linear relationships, making it suitable for complex biological datasets like those found in milk quality classification.

SVM has seen extensive application in the food industry, particularly in quality classification, adulteration detection, and safety monitoring. For milk quality assessment, SVM models have been employed to classify samples based on physicochemical parameters such as pH, fat content, temperature, and sensory attributes like taste and odor. In order to forecast milk quality based on seven characteristics pH, temperature, taste, odor, fat content, turbidity, and color [5] used support vector machines (SVM). The 1,059 milk samples in their dataset, which they obtained via Kaggle, were categorized as poor, medium, or high quality. The SVM model outperformed conventional statistical classifiers with an accuracy of 91.6%. Applying kernel tricks, particularly the radial basis function (RBF), which enabled non-linear class separation, improved the model. In another study, [19] applied SVM to classify milk samples based on data obtained using electronic sensors (E-nose), achieving classification accuracies over 90%. The ability of SVM to generalize from small sample sizes and handle high-dimensional data makes it particularly effective for food quality prediction

In a thorough mapping study on machine learning applications in dairy farms, [17] found that SVM consistently ranked among the best algorithms for predicting the quality of milk output, feed efficiency, and health status of dairy animals. Their research focused on SVM's capacity to handle small and noisy datasets, which is a typical problem in developing nations.

In the context of milk quality, the features commonly used in SVM models include Physicochemical parameters: pH, fat content, protein, temperature, Microbiological indicators: Total bacterial count, somatic cell count, Sensory data: Taste, odor, color, Environmental variables: Ambient temperature, storage duration.

The Kaggle milk quality dataset, used in several studies including [5] and [9], is a popular benchmark dataset that includes seven numerical features. However, this dataset lacks microbiological data, which limits its use in comprehensive quality assessments.

In contrast, [14] used a richer dataset obtained from 10 farms in China, which included Dairy Herd Improvement (DHI) data and E-nose sensor readings.

These datasets were preprocessed using standard techniques such as normalization, feature scaling, and principal component analysis (PCA). PCA was particularly helpful in reducing dimensionality before feeding data into the SVM model, which is sensitive to multicollinearity among features.

SVM models are evaluated using accuracy as a proportion of correct predictions, precision and Recall that are particularly important in detecting poor-quality milk that has a minority class, F1-score that gives the harmonic mean of precision and recall, and ROC-AUC to evaluate classification performance across all thresholds.

While SVM has proven performance in controlled datasets, its applicability in Uganda is both promising and constrained. SVM's robustness in small datasets and noisy environments is advantageous, considering the limited and sometimes inconsistent nature of data in Ugandan milk processing plants.

However, challenges remain: Data Requirements: SVM models still require labeled training data, which is sparse in Uganda; Computational Complexity: SVMs are computationally intensive, particularly with large datasets or when using non-linear kernels; Interpretability: Unlike decision tree-based models, SVM is a "black box" model less intuitive for plant operators and policymakers.

Despite these challenges, SVM offers a viable route for early adoption in Uganda's dairy sector if coupled with proper data preprocessing and supported by user-friendly interfaces. It can serve as a baseline model or be integrated into hybrid systems with more interpretable models.

2.3.3 Random Forest (RF) in Milk Quality Classification

Table 1: Strengths and Weaknesses of SVM

Strengths	Weaknesses
High accuracy in classifying milk quality.	Requires careful parameter tuning (C, gamma).
Handles high-dimensional and non-linear data well.	Computationally expensive for large datasets.
Robust to overfitting when regularized.	Difficult to interpret (black-box nature).

Strengths	Weaknesses
Effective even with relatively small and noisy datasets.	Performance heavily depends on proper kernel selection.
Well-supported in libraries like Scikitlearn and easy to integrate in Python.	Needs numerical input; categorical variables must be encoded in advance.

Random Forest (RF) is an ensemble machine learning method introduced by Breiman (2001) that operates by constructing a multitude of decision trees during training and outputting either the mode of the classes (classification) or the mean prediction (regression) of the individual trees. Each tree in the forest is trained on a bootstrap sample of the data, and a random subset of features is considered when splitting nodes. This "bagging" approach helps reduce variance and improves generalization, making Random Forest highly robust to noise and overfitting.

RF has emerged as one of the most reliable algorithms in the field of food quality prediction due to its simplicity, adaptability to high-dimensional datasets, and strong predictive performance. It's interpretability through feature importance scores also makes it valuable for decision support in practical settings like milk processing plants

Random Forest has been widely adopted in dairy research, particularly for quality classification, microbial contamination detection, and traceability.

In order to predict milk quality, [9] used a dataset that included seven physicochemical and sensory factors to compare Random Forest and Support Vector Machines. At 94.2% accuracy, the RF model marginally outperformed the SVM (91.6%). To manage multicollinearity and capture intricate relationships between input variables, the authors emphasized RF's capabilities.

In a more advanced implementation, [14] developed RF models to estimate milk fat and protein content using data from electronic nose (E-nose) sensors and Dairy Herd Improvement (DHI) laboratory results from 1,000 milk samples in China. The RF model achieved a high coefficient of determination ($R^2 = 0.93$) and low root mean squared error (RMSE), indicating excellent predictive power. This study is particularly relevant for Uganda, where similar multidimensional datasets (chemical, microbial, and sensory) could be used in future implementations

Datasets and Feature Sets RF models for milk quality classification generally utilize the following features: Chemical features: pH, fat content, protein, lactose, total solids, Microbiological indicators: Somatic cell count (SCC), total bacterial count, Sensor-based data: E-nose readings, turbidity, and temperature fluctuations; Sensory attributes: Odor, taste, color.

The 1,059-sample Kaggle milk quality dataset was used by [5] and [9]. By using a specific dataset that included both sensor data and laboratory test results, [14] were able to apply regression-based RF models for quantitative prediction as opposed to only categorization.

In each case, feature selection and dimensionality reduction were critical. PCA was used to identify the most significant predictors, and RF's internal feature importance metrics further validated the selected features.

RF models are typically assessed using accuracy that gives overall correctness of classification, precision/recall/F1-score for imbalanced datasets, mean absolute error (MAE) and root mean squared error (RMSE) for regression tasks, and R^2 (coefficient of determination) to evaluate explained variance in regression models. These metrics consistently place Random Forest among the top-performing models for both classification and regression tasks in milk quality analytics.

Random Forest is particularly suited to the Ugandan dairy context for several reasons:

RF provides feature importance scores, which can help processors and regulators understand what factors most affect milk quality, an essential trait in environments where trust in automated systems may be limited.

Robustness to Noise and Missing Data: RF models are less sensitive to noise and partial data loss, making them ideal for datasets with gaps common in Ugandan milk collection centers where manual record-keeping is still prevalent.

RF models can handle large datasets without drastic increases in computational cost. This allows scalability as data infrastructure improves in Uganda.

Minimal Preprocessing Requirements: Unlike SVM or neural networks, RF does not require data normalization or transformation, simplifying implementation in resource-constrained settings.

However, there are some limitations with RF models; they can become computationally heavy with very large datasets or when many trees are used. Overfitting still occurs if the number of trees and depth are not well regulated. While more interpretable than neural networks, RFs still fall short of the transparency offered by decision trees.

Nonetheless, the algorithm's balance of accuracy, robustness, and partial interpretability makes it one of the most promising choices for early deployment in Uganda's dairy sector.

Table 2: Strengths and Weaknesses of Random Forest (RF)

Strengths	Weaknesses
High accuracy and strong generalization capability.	Can be computationally intensive for large ensembles.
Handles multicollinearity and highdimensional data well.	Less transparent than single decision trees.
Built-in feature importance aids in interpretability.	May overfit if hyperparameters are not tuned.
Robust to missing data and noise.	Performance can degrade with highly imbalanced datasets.
No need for data normalization or scaling.	Needs careful selection of number of trees and depth to avoid redundancy.

2.3.4 Comparative Summary of Machine Learning Models for Milk

Quality Prediction

Artificial Neural Networks (ANNs) are computational models inspired by the structure and functionality of the human brain. They consist of interconnected layers of nodes (neurons), where each connection has an associated weight adjusted through training. ANNs are particularly effective in modeling complex, non-linear relationships, making them suitable for pattern recognition and prediction tasks in diverse fields, including food quality assessment and process control.

ANNs are classified into various architectures such as feedforward neural networks (FNN), convolutional neural networks (CNN), and recurrent neural networks (RNN). In milk quality classification, feedforward backpropagation networks are the most commonly used, where data flows from the input layer through one or more hidden layers to the output.

Applications in Dairy and Food Quality Assessment

ANNs have been increasingly adopted for milk quality classification, contamination detection, adulteration analysis, and prediction of chemical composition. Their ability to approximate any function and capture interactions among multiple input features makes them highly adaptable for complex quality metrics.

In order to categorize milk samples as low, medium, or high quality based on at [19] created artificial neural network (ANN) models using the Orange Data Mining Tool. One hundred samples per class were used to train the ANN model, and cross-validation was used to confirm its accuracy. Although its classification accuracy of 88.5% was marginally less than that of AdaBoost (91.2%) in the same trial, it showed greater flexibility when dealing with noisy input.

Similarly, [4] reviewed over 50 applications of ANN in the food industry and concluded that ANNs consistently ranked among the top three performers in tasks such as shelf-life prediction, microbial growth estimation, and physicochemical quality control. In milk processing, ANN models have been applied to predict somatic cell count, fat content, protein concentration, and detection of microbial spoilage indicators.

In order to estimate milk fat and protein contents, [14] also investigated the use of ANN in combination with sensor data. The ANN showed good performance on complicated datasets, including sensor fusion, even though their Random Forest model performed marginally better than the ANN in terms of R^2 and MAE.

Feature Engineering and Datasets for Milk quality prediction using ANN typically involve input features such as Chemical attributes: pH, fat content, protein, lactose, total solids, Sensory attributes: Taste, odor, color, texture, Environmental conditions: Storage temperature, duration, humidity, sensor data: E-nose readings, turbidity, and real-time temperature logs.

Since the ANN model could naturally accommodate non-linear interactions, all seven features from the Kaggle dataset were fed into it in [19] experiment without dimensionality reduction. In more sophisticated implementations, sensor data was preprocessed using PCA before being fed into the ANN in order to decrease noise and accelerate convergence [14].

Unlike simpler algorithms, ANN models require careful feature scaling (normalization) to ensure effective weight updates during backpropagation. This preprocessing step is crucial in improving training speed and predictive performance.

ANNs are evaluated using standard regression and classification metrics, depending on the application: Classification metrics: Accuracy, precision, recall, F1-score, confusion matrix.

Regression metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2.

ANN performance can be significantly improved through hyperparameter tuning: adjusting learning rate, batch size, number of epochs, number of hidden layers, and neurons. Regularization techniques: Dropout and L2 regularization to reduce overfitting.

Cross-validation: Typically k-fold to ensure generalizability.

While the results slightly lagged behind ensemble models like AdaBoost in classification tasks, ANN models provided superior performance in regression problems involving continuous variables such as fat and protein content.

The application of ANN in the Ugandan dairy sector is both promising and challenging. Its strength in capturing non-linear dependencies is ideal for heterogeneous datasets typical of smallholder milk production systems where quality is affected by inconsistent handling, variable feeds, and seasonal effects.

Table 3: Strengths and Weaknesses of Artificial Neural Networks (ANN)

Strengths	Weaknesses
Capable of modeling complex non-linear relationships.	Requires large, labeled datasets for training.
Handles multi-dimensional and noisy data well.	High computational requirements for training and inference.
Adaptable to both classification and regression tasks.	Black-box model limited interpretability.
Can integrate data from multiple sources and modalities.	Sensitive to hyperparameter selection and overfitting.
Improves with data scalable as more quality records are digitized.	Needs data normalization and tuning to perform optimally.

2.3.5 K-Nearest Neighbors (KNN and Distance-Weighted KNN) in Milk Quality Classification

The K-Nearest Neighbors (KNN) algorithm is a non-parametric, instance-based learning model used for classification and regression. It operates on the principle that similar instances exist in close proximity in feature space. For a given input, the algorithm identifies the k closest training samples based on a distance metric known as Euclidean distance and predicts the label or value based on the majority vote classification or average regression of the neighbors.

An extension of the standard KNN is Distance-Weighted KNN (DW-KNN), where the influence of each neighbor is weighted by its inverse distance to the input point. This improves prediction accuracy, especially in noisy datasets or when class boundaries are irregular. Due to its simplicity, intuitive logic, and effectiveness in various domains, KNN has found applications in food quality assessment, including milk quality classification.

KNN has been applied in Dairy and Food Quality Prediction using a Kaggle dataset, [15] compared the effectiveness of ordinary KNN and DW-KNN for classifying milk quality.

1,059 milk samples that were classified as poor, medium, or high quality according to characteristics such as pH, temperature, turbidity, fat content, taste, odor, and color made up the dataset. With a classification accuracy of 89.7% versus 84.2% for normal KNN, the researchers showed that DW-KNN continuously beat traditional KNN, even if the latter had reasonable results.

The enhancement was attributed to DW-KNN's ability to assign higher weights to nearer samples, effectively emphasizing more relevant data points during classification. This is especially valuable in milk datasets where subtle chemical or sensory differences can determine quality class.

In their experimentation with KNN and other models, [5] observed that although KNN was easy to implement, its overall accuracy and resilience were not as good as those of more sophisticated models like Random Forest and SVM. However, its performance was competitive enough to justify its use in low-resource or real-time contexts where interpretability and ease of use are critical.

Common features used for KNN-based milk quality classification included physicochemical attributes such as pH, temperature, turbidity, fat, and protein; sensory attributes like taste, odor, and color; and temporal/environmental conditions such as storage time and handling practices.

The Kaggle dataset used by [15] and [19] provided a practical foundation for KNN model development. No complex preprocessing or transformation such as PCA was necessary, though data normalization was critical due to the algorithm's sensitivity to feature scales. Features were normalized to a common range using min-max scaling or z-score standardization to ensure fair distance computations.

Nonetheless, KNN and DW-KNN also present several practical challenges in the Ugandan context such as scalability; As more data is collected over time, prediction time increases due to the need to compute distances to all stored instances and data storage needs. The full dataset must be retained, which could be burdensome without stable digital infrastructure and sensitivity to noisy and irrelevant features. Without proper feature selection or dimensionality reduction, model accuracy suffers and requires feature normalization, a preprocessing step not always feasible in nonautomated environments.

While KNN alone may not be the optimal model for comprehensive milk quality classification, it can serve as a baseline model or a component within a hybrid decision support system in Uganda's milk processing plants.

Table 4: Strengths and Weaknesses of KNN and DW-KNN

Strengths	Weaknesses
Simple to understand and implement.	Computationally expensive for large datasets (lazy learning).
Effective with small datasets.	Requires full data storage and fast retrieval systems.
No training time; useful in dynamic or streaming data scenarios.	Performance highly sensitive to irrelevant/noisy features.
DW-KNN improves prediction by emphasizing closer data points.	Needs proper feature scaling/normalization.
Suitable for real-time quality alerts in low-data environments.	Poor performance on high-dimensional or imbalanced datasets.

2.3.6 Gradient Boosting Machines (GBM) and XGBoost in Milk Quality classification

Gradient Boosting Machines (GBMs) are powerful ensemble learning techniques that combine the predictions of multiple weak learners, typically decision trees, in a sequential manner. Each tree corrects the errors made by the previous ones by fitting to the residuals (errors) of the prediction, gradually improving performance. Introduced by Friedman (2001), GBM has become a core algorithm for structured data problems due to its flexibility, high predictive accuracy, and ability to handle various loss functions. One of the most well-known implementations of GBM is Extreme Gradient Boosting (XGBoost). Developed by [7], XGBoost introduced several innovations, including regularization (L1 and L2), tree pruning, parallelized computation, and optimized handling of missing values, all of which significantly improve speed and performance. These features make XGBoost a preferred model in data science competitions and industrial applications, including agriculture and food safety.

Applications in Milk and Food Quality Assessment

GBM and its variants, XGBoost and LightGBM, have shown high effectiveness in food quality prediction tasks, particularly when dealing with heterogeneous datasets or when capturing interactions between physicochemical and environmental variables is crucial.
Using sensor data and Dairy Herd Improvement (DHI) lab findings gathered from ten Chinese cattle farms, [14] employed Gradient Boosting Decision Trees (GBDT) and XGBoost to estimate milk fat and protein content. Their models were successful in:

$R^2 = 0.94$ for fat prediction

< 0.12

< 0.19

XGBoost outperformed all other tested models, including Support Vector Machines (SVM) and Artificial Neural Networks (ANN), particularly in handling multicollinearity and outliers.

The effectiveness of GBM in forecasting deterioration, contamination concerns, and shelf-life in meat and dairy products was emphasized by [4]. In complex food datasets, the model's ability to rank and choose significant features was also helpful in locating crucial quality markers. In a comparative study, [5] tested XGBoost alongside Random Forest and SVM in milk quality classification. Although RF had slightly higher precision, XGBoost achieved the best balance across accuracy (95.1%), recall (94.5%), and F1-score (94.7%), affirming its strength in handling imbalanced datasets and complex feature interactions.

GBM models work well with raw tabular data like pH, turbidity, fat, temperature, taste, odor, and color; sensor data such as volatile compounds from electronic noses, digital thermometers, and humidity sensors.

In [14], feature selection using XGBoost's importance metrics helped identify fat and protein predictors such as total solids, SCC, temperature, and specific sensor voltages. XGBoost's in-built regularization also reduces the need for aggressive preprocessing like normalization or PCA, unlike SVM or ANN.

Performance metrics for GBM and XGBoost models typically include: Accuracy, precision, recall, and F1-score for classification tasks; MAE, RMSE, and R^2 for regression tasks; log loss and AUC-ROC for probabilistic classification and imbalanced data handling.

The following was reported by [5]:

Accuracy: 95.1%

Precision: 0.94

Recall: 0.945

F1-score: 0.947

AUC-ROC: 0.96

Gradient Boosting, especially in the form of XGBoost, offers tremendous potential for deployment in Uganda's milk processing plants with the following key advantages: High predictive power, which is suitable for noisy and sparse datasets prevalent in Ugandan settings. It also handles diverse data types such as environmental, microbial, chemical, and sensory, and it has built-in feature selection and missing value handling techniques that reduce preprocessing burden.

XGBoost models present challenges in Uganda, like computational intensity, where training XGBoost models can be demanding, requiring hardware and skilled operators not readily available in rural areas. Also, these models need careful tuning because suboptimal parameters can lead to overfitting, underfitting, or excessive computation.

Despite these challenges, the trade-off between accuracy and explainability can be managed by coupling XGBoost with visual dashboards, post-hoc explanation tools such as SHAP values, and decision support layers. With modest investments in infrastructure and training, GBM-based models could provide high-accuracy predictive tools for quality control in Uganda's dairy value chain.

Table 5: Strengths and Weaknesses of GBM/XGBoost

Strengths	Weaknesses
Excellent predictive accuracy in both classification and regression.	Requires substantial computational resources for training.
Handles missing data and categorical features effectively.	Model interpretability is limited.
Automatically identifies important predictors.	Sensitive to overfitting if not properly regularized.
Performs well on small to medium-sized structured datasets.	Hyperparameter tuning is complex and time-consuming.
Strong performance in imbalanced and noisy datasets.	May not be feasible in low-resource environments without infrastructure.

2.3.7 Sensor-Integrated Machine Learning Models: Electronic Nose (E-Nose) with ML Algorithms

Sensor-integrated machine learning models represent a modern fusion of hardware-based sensing systems and data-driven predictive algorithms. In dairy and food industries, Electronic Nose (E-Nose) systems simulate the human olfactory system by using sensor arrays to detect volatile compounds. These signals are transformed into electrical responses, which are then processed using machine learning (ML) techniques for classification or regression purposes.

An E-Nose typically comprises an array of chemical sensors, such as metal oxide semiconductors and quartz crystal microbalances, a data acquisition system, and a pattern recognition unit, where ML algorithms like SVM, Random Forest, and ANN interpret sensor outputs.

The combination enables real-time, non-destructive milk quality evaluation, ideal for detecting spoilage, contamination, or compositional deviations that traditional methods might miss or take too long to identify.

One of the most thorough research studies on the application of E-Nose technology in conjunction with machine learning algorithms to forecast milk fat and protein content was carried out by [14]. Their setup for the experiment included collecting 1,000 raw milk samples from 10 dairy farms, using a seven-sensor E-Nose system to measure volatile compounds, and applying Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) to reduce data dimensionality.

Feeding the processed data into machine learning models, specifically Support Vector Machines (SVM), Random Forest (RF), and Gradient Boosting Machines (GBM). The Random Forest model trained on fusion features (combined sensor + lab data) performed best, achieving:

$R^2 = 0.93$ for fat prediction.

MAE = 0.11

RMSE = 0.16

The fusion approach greatly enhanced accuracy compared to using lab data or E-Nose data alone. The study demonstrated the viability of integrating sensors with ML for milk quality estimation in near real-time.

Additionally, [19] used ANN and Adaptive Boosting (AdaBoost) to explore with sensor-driven models. Milk samples were manually assessed and run through sensor arrays, and quality classification predictions were established. Despite the ANN model's good performance (88.5% accuracy), AdaBoost beat it (91.2%).

E-Nose systems produce multivariate time-series or voltage-based data depending on the sensor array. The dataset consisted of sensor readings like voltage outputs from each sensor in the array, derived feature like maximum sensor response, area under curve, response time, recovery time, Lab reference values like pH, fat %, protein %, lactose, bacterial load, etc.

In order to provide multi-modal ML input, [14] integrated E-Nose data with DHI data (fat, protein, solids, and SCC).

In [14], sensor-integrated models consistently outperformed standalone ML models using only tabular features. Models with sensor fusion reduced false positives in quality classification, improved early detection of low-fat or adulterated samples, and enabled regression predictions of fat/protein with R^2 above 0.90.

The integration of E-Nose sensors with machine learning offers immense potential for Uganda's milk processing sector, particularly in real-time spoilage detection since it's faster than laboratory analysis, enabling proactive quality rejection.

Table 6: Strengths and Weaknesses of E-Nose + ML Models

Strengths	Weaknesses
Enables real-time, non-destructive milk quality evaluation.	High initial equipment and calibration costs.
Detects spoilage and adulteration faster than lab methods.	Requires specialized maintenance and trained personnel.
Suitable for remote areas without laboratory infrastructure.	Environmental factors can affect sensor readings.
Enhances ML performance via sensor-lab data fusion.	Integration with existing systems may be complex.
Facilitates predictive modeling and early intervention.	Data preprocessing and dimensionality reduction are required.

2.3.8 Comparative Summary of Machine Learning Models for Milk Quality Classification

To evaluate and compare the six machine learning models discussed in Sections 2.2.1 to 2.2.6, the table below summarizes their core strengths, limitations, performance trends, and contextual suitability for the Ugandan dairy industry.

Table 7: Comparative Summary of ML Models for Milk Quality Classification

Model	Strengths	Weaknesses	Performance Metrics	Suitability for Uganda
SVM (Support Vector Machines)	High accuracy, handles nonlinear and high-dimensional data well, performs with small datasets.	Requires careful tuning, high computational cost, and a black-box model.	Accuracy ≈ 91.6%, F1 ≈ 0.91	Suitable if data is limited; may need interface and explanation tools.
RF (Random Forest)	Excellent accuracy, robust to noise, provides feature importance, easy to tune.	May overfit with deep trees and be less interpretable than single trees.	Accuracy ≈ 94.2%, $R^2 \approx 0.93$	Highly suitable; handles incomplete or uncleaned datasets well.
ANN (Artificial Neural Networks)	Captures complex relationships, adapts to various data types,	Requires large datasets, slow to train, low interpretability.	Accuracy ≈ 88.5%, RMSE < 0.2	Moderate suitability; powerful but needs

Model	Strengths	Weaknesses	Performance Metrics	Suitability for Uganda
	powerful in regression tasks.			infrastructure and training.
KNN / DW-KNN	Simple, interpretable, no training phase, performs well on small data.	Inefficient at scale, sensitive to noise and irrelevant features.	Accuracy ≈ 84%–89.7%	Suitable for small-scale, low-tech deployments; not scalable long-term.
GBM / XG-Boost	State-of-the-art performance, handles missing values and categorical features, and feature selection built-in.	Black-box, computationally heavy, complex tuning required.	Accuracy ≈ 95.1%, $R^2 \approx 0.94$	Highly suitable where accuracy is key and digital infrastructure exists.
Model	Strengths	Weaknesses	Performance Metrics	Suitability for Uganda
	categorical features, feature selection built-in.			
E-Nose + ML	Real-time, non-destructive, objective testing fuses lab and sensor data for better prediction.	Requires specialized hardware, training, calibration, and integration.	$R^2 \approx 0.93$, MAE < 0.12	Very promising; ideal for mobile/automated quality monitoring units.

2.3.9 Model Selection and Justification

From the review above, it is evident that different ML models offer unique advantages and drawbacks, with varying levels of feasibility depending on technical, infrastructural, and operational constraints.

SVM is powerful for small, clean datasets but is less interpretable and resource-demanding.

Random Forest provides a strong balance between accuracy and ease of deployment, and it handles messy, real-world data well.

ANN is flexible and powerful but requires extensive data and tuning, making it less ideal for resource-constrained environments.

KNN is ideal for smaller, less complex environments but lacks scalability and can underperform with large or noisy datasets.

XGBoost offers best-in-class performance but demands significant computing resources and expertise.
E-Nose with ML represents the most advanced approach, merging hardware and AI for real-time prediction, but it requires sensor investments and is still emerging in developing countries.

2.3.10 Model Selection for the Current Study

After reviewing and comparing the models discussed in this chapter, Random Forest was selected as the principal model for the current study. This decision was based on its strong reported classification performance, robustness when handling structured tabular datasets, and suitability for deployment in a practical prototype system.
Random Forest also offers an important interpretability advantage through feature-importance ranking, which is valuable in quality-control contexts where plant operators and supervisors need to understand the variables influencing classification outcomes. In addition, the algorithm requires relatively limited preprocessing compared with some alternative models and is more practical for implementation on standard desktop hardware.
Although other models such as SVM, ANN, and XGBoost were considered and evaluated for comparative purposes, the current study prioritized Random Forest as the final deployment model. The prototype system was therefore aligned to the use of Random Forest for milk quality classification while retaining a comparative discussion of other machine learning techniques within the broader literature review.

2.4 Decision Support Systems and Prototype Systems in Milk Quality Management

Decision Support Systems (DSS) are computer-based systems designed to assist users in making informed, consistent, and timely decisions through the analysis of data, models, and domain knowledge. In agricultural and dairy environments, DSS have increasingly been adopted to improve operational efficiency, support quality control, enhance disease surveillance, and optimize resource utilization. Existing literature shows that dairy-related DSS are already being applied in areas such as herd management, feed planning, disease detection, and performance monitoring. These systems demonstrate that data-driven decision support is not a new concept in dairy production; rather, it is an established approach with proven value in transforming large volumes of structured data into actionable management insights. However, most of these systems have largely focused on farm productivity and animal management, with less emphasis on plant-level milk quality classification and operational decision support within dairy processing environments [1].

In the context of milk quality management, decision support approaches have evolved from conventional reactive inspection methods toward more proactive and intelligent systems. One notable contribution is the early detection system proposed for raw milk quality management in Australia, which integrated sensor monitoring, rule-based logic, and machine learning techniques to support proactive decision-making. That system monitored critical indicators such as milk temperature and storage conditions in order to detect quality risks before milk deterioration became severe. The significance of this approach lies in its shift from symptom-based inspection to preventive quality management, allowing timely intervention before the milk proceeds further along the supply chain. This demonstrates that prototype-oriented decision support systems can function effectively when real-time data acquisition and predictive logic are combined in a practical framework [2].

Similarly, recent prototype systems have shown that automated and intelligent milk quality assessment is technically feasible and operationally beneficial. For example, a portable robotic arm system developed for raw milk quality assessment integrated multiple sensors, including pH, total dissolved solids, temperature, density, and color, with an artificial intelligence classifier for real-time milk freshness classification. The study reported strong classification performance, high repeatability, and a reduction in testing time when compared with manual procedures. Such findings indicate that prototype systems are already capable of supporting rapid, data-driven milk quality evaluation and can potentially reduce human subjectivity in quality inspection. This is particularly important in dairy settings where timely assessment is critical for maintaining product safety and reducing post-harvest losses [3].

Despite these advances, a considerable portion of the literature remains concentrated on predictive model performance rather than on the development of deployable end-user systems. Several studies on milk quality classification have demonstrated that machine learning algorithms can achieve high levels of predictive accuracy using physicochemical features such as pH, temperature, taste, odor, turbidity, fat, and color. While these studies provide strong evidence that classification models can distinguish between low-, medium-, and high-quality milk, many of them stop at experimental validation and do not extend their work into practical prototype development. As a result, the analytical model may perform well in a research environment, yet the absence of a user-oriented interface limits its value in real operational settings where dairy personnel require simple, interpretable, and accessible decision support tools [4].

Additional evidence from digital dairy quality initiatives further confirms that decision support concepts are being used in practical quality management applications. For instance, the Milk Quality Improvement Program at Cornell University has documented the development of predictive and digital tools for spoilage modelling, contamination analysis, and quality management in dairy systems. These contributions illustrate that digital tools can support specialized quality-related decisions in dairy industries and can enhance preventive management of spoilage risks. Nevertheless, such systems are often tailored to specific quality problems, advanced industrial conditions, or highly specialized operational contexts, which may reduce their direct applicability in low-resource processing environments where infrastructure, technical expertise, and system accessibility remain constrained [5].

The literature also reveals several shortcomings in existing DSS and prototype systems. First, many systems remain reactive in nature, meaning that they identify quality deterioration after it has already occurred or after milk has advanced to later stages of the supply chain. This delays corrective action and may lead to rejection, waste, and financial loss. Second, some systems depend on stable electricity supply, controlled environmental conditions, calibrated hardware, or advanced computing infrastructure, thereby limiting their suitability for rural or resource-constrained dairy settings. Third, a number of decision support models have been criticized for limited user-friendliness, weak integration with everyday workflows, and insufficient accessibility for non-technical users. Fourth, some prototype studies have been validated on relatively small datasets or within narrowly defined geographical contexts, making it difficult to generalize their effectiveness across different production and processing environments. These weaknesses suggest that technical functionality alone is insufficient unless accompanied by usability, contextual relevance, and operational feasibility [1], [2], [3], [6].

From the foregoing review, it is evident that existing studies have made important progress in demonstrating that decision support systems, sensor-enabled monitoring platforms, and machine learning models can improve milk quality assessment. What is currently working includes automated sensing, predictive classification, spoilage forecasting, and the use of prototype systems to reduce manual subjectivity and improve timeliness of decisions. However, what remains inadequate is the translation of these technical advances into simple, interpretable, affordable, and context-appropriate operational tools for routine use in dairy processing plants, especially in developing-country settings. In Uganda, where dairy processors often operate under infrastructural limitations such as intermittent electricity, limited digital integration, and continued reliance on manual quality checks, there remains a practical gap in the development of a user-oriented prototype system that can support consistent milk quality classification and guide routine quality-related decisions. It is this gap that the current study seeks to address through the application of a Random Forest model integrated into a prototype system for milk quality classification in Ugandan dairy processing plants [1], [3], [4].

Chapter Three

Methodology

This chapter details the systematic procedures, tools, and frameworks used to achieve the objectives of this study. It outlines the research design, dataset synthesis based on comprehensive regional standards, machine learning model development, software engineering architecture for the DairyIQ prototype, and the system validation framework for classifying milk quality in milk processing plants in Uganda.

3.1 Data Collection

3.1.1 Data Sources

The utilization of a synthesized dataset bounded by the East African Standard (EAS 67:2019) serves as a legally grounded, scientifically objective approach to overcome the data accessibility barriers common in private industrial sectors. In the Ugandan dairy industry, historical milk quality records are treated as highly sensitive, proprietary business intelligence. Consequently, local processing plants routinely withhold these metrics to safeguard commercial privacy and protect supply chain relationships. To maintain structural neutrality while ensuring strict compliance with local regulatory frameworks, the DairyIQ predictive engine was trained on a parameter matrix strictly controlled by regional statutory thresholds rather than localized, biased facility data. Furthermore, while real-world milk reception typically exhibits a highly skewed distribution dominated by passing grades, a deliberately balanced allocation of 1,500 samples (500 entries per target quality tier) was generated. In machine learning architectures, highly imbalanced datasets introduce a severe majority-class training bias, causing models to overlook rare but critical failure states. By employing an evenly distributed parametric matrix, the Random Forest classifier is forced to give equal statistical weight to all quality conditions. This design significantly enhanced the sensitivity of the DairyIQ software system, ensuring it reliably detect contaminated, adulterated, or borderline-moderate milk deliveries at the processing dock.

3.1.2 Data Collection Methods

To ensure commercial neutrality, legality, and direct relevance to regional dairy enforcement, training data for the model was synthesized based on the threshold distributions stipulated in the East African Standard (EAS 67:2019) for Raw Cow Milk Specifications. A balanced dataset of 1,500 simulated milk samples (500 per target category) was mathematically generated using Python's numpy.random module.

The dataset includes all 11 critical physical, chemical, organoleptic, and bacteriological features audited at raw milk reception docks:

1. Physicochemical & Compositional: pH, **temperature (C)**, fat content (%), titratable acidity (% lactic acid), protein content (%), and lactose content (%).
2. Microbiological & Cellular: Total Plate Count (TPC, CFU/ml) and Somatic Cell Count (SCC, cells/ml).
3. Organoleptic (Sensory): Taste, odor, and color.

The target output is modeled as a multi-class variable categorizing milk quality as High, Moderate, or Low based on regional quality tiers and penalization zones.

Table 3.1: Parametric Matrix Framework for Class Synthesization

For sensory metrics (taste, odor, and color), inputs are mapped to integers (0) for off/abnormal and (1) for normal.

Feature Parameter	Normal EAS 69 Limits	High Quality (N=500)	Moderate Quality (N=500)	Low Quality (N=500)
pH	$6.6 \leq pH \leq 6.8$	6.65 - 6.75	6.60 - 6.64 or 6.76 - 6.80	<6.60 (Sour) or >6.80

Temperature (°C)	≤10.0°C	2.0°C-6.0°C	6.1°C-10.0°C	>10.0°C (Chain break)
Fat Content (%)	3.25%	3.80%-5.50%	3.25%-3.79%	<3.25% (Watered/Skimmed)
Titratable Acidity (%)	0.17	0.12%-0.14%	0.15%-0.17%	>0.17% (Souring lactic acid)
Protein Content (%)	3.20%	3.40%-4.50%	3.20%-3.39%	<3.20% (Adulteration/Poor feed)
Lactose Content (%)	4.40%-4.90%)	4.60%-4.85%	4.40%-4.59%	<4.40% (Mastitis depletion)
Total Plate Count (CFU/ml)	≤2.0×10^5	<5.0×10^4	5.0×10^4 -2.0×10^5	>2.0×10^5 (High bacteria)
Somatic Cell Count (cells/ml)	≤5.0×10^5	<2.0×10^5	2.0×10^5-5.0×10^5	>5.0×10^5 (Mastitis present)
Taste	Normal(1)	1 (Normal)	1 (Normal)	0 (Off-flavour/Tainted)
Odor	Normal(1)	1 (Normal)	1 (Normal)	0 (Sour/Objectionable)

Color	Creamy White (1)	1 (Normal)	1 (Normal)	0 (Abnormal/Bloody/Yellow)

3.2 Data Preprocessing

3.3 Data Pre-processing and Feature Engineering

Before deployment within the machine learning pipeline, the multiparametric data underwent three primary preprocessing operations:

3.3.1 Label and Categorical Encoding

The nominal quality targets were mapped into ordinal numerical representations for optimization:

High 0, Moderate 1, Low 2

Sensory features input as dropdowns by the laboratory operator (e.g., "Normal" and "Off-Flavor") were transformed into binary integer indicators, 1 and 0, respectively, within the application's mapping layer.

3.3.2 Logarithmic Transformation

Microbiological parameters exhibiting highly skewed distributions, specifically Total Plate Count (TPC) and Somatic Cell Count (SCC), were subjected to a base-10 logarithmic transformation:

$X_{log} = \log10(X + 1)$

This transformation standardizes the numerical variance, ensuring the tree splits are not overwhelmed by large absolute scales.

3.3.3 Dataset Partitioning

The processed, 11-feature dataset was randomized and split into independent partitions using an (80:20) ratio:

1. Training Subsample (80%/1200 records): Used by the algorithm to learn decision paths.
2. Testing Subsample (20%/300 records): Held back entirely to check generalization metrics against unseen data configurations.

3.4 Model Selection and Training (Random Forest Classifier)

The core predictive engine uses a Random Forest Classifier, an ensemble machine learning algorithm that constructs an iterative forest of individual decision trees during the training phase. This specific model was chosen for the DairyIQ prototype because:

1. It natively supports multi-class classification assignments without requiring complex data binarization setups.
2. It accommodates mixed-type data features simultaneously, processing continuous chemical records alongside binary sensory inputs.
3. It measures feature importance weights, allowing researchers to evaluate which test parameter contributes most heavily to a "Low" quality designation.

The model uses bootstrap aggregation (bagging) to create subsets of data, training individual decision trees **Ĉk(x)** on each subset. The ultimate classification score for an input vector \(x\) containing the 11 lab parameters is calculated via a majority vote across all \(B\) built trees:

\(\^{C}_{rf}^{B}(x)=\text{arg\ max}_{c\in \{\text{High,\ Moderate,\ Low}\}}\sum _{k=1}^{B}I\left(\^{C}_{k}(x)=c\right)\)

During architectural compilation, hyperparameters were pinned to \(n_estimators=100\) and \(max_depth=12\) to protect the engine from localized noise overfitting while ensuring immediate runtimes on factory workstations.

3.5 Model Evaluation Metrics [1]

Because this study addresses a three-class classification problem, model performance was evaluated using macro-averaged evaluation metrics calculated from a multi-class confusion matrix tracking True Positives (TP_{c}), False Positives (FP_{c}), and False Negatives (FN_{c}) for each respective class $c \in \{0, 1, 2\}$:

3.5.1 Accuracy [1]

Measures the overall global percentage of correct quality predictions across all test samples:

$$\text{Accuracy}=\frac{\sum _{c}\text{Correct\ Classifications}_{c}}{\text{Total\ Tested\ Samples}}$$

3.5.2 Macro-Precision

Calculates the precise reliability of the quality classifications averaged across the three tiers:

$$\text{Precision}_{\text{macro}}=\frac{1}{3}\sum _{c=0}^{2}\frac{TP_{c}}{TP_{c}+FP_{c}}$$

3.5.3 Macro-Recall

Measures the system's sensitivity to catching problematic or moderate milk batches across all incoming deliveries:

$$\text{Recall}_{\text{macro}}=\frac{1}{3}\sum _{c=0}^{2}\frac{TP_{c}}{TP_{c}+FN_{c}}$$

3.5.4 Macro-F1-Score

Calculates the balanced performance metric, capturing the balance between precision and recall across all target classes:

$$\text{F1-Score}_{\text{macro}}=2\times \frac{\text{Precision}_{\text{macro}}\times \text{Recall}_{\text{macro}}}{\text{Precision}_{\text{macro}}+\text{Recall}_{\text{macro}}}$$

3.2.1 Data Cleaning

Data cleaning ensured data quality and completeness through the following measures: - Missing data points were handled using imputation techniques to retain data integrity. Interpolation estimated missing values based on neighboring data points for a smooth dataset. Mean or median imputation handled non-sequential missing values, replacing empty values with the median or mean. These techniques ensured data completeness, reducing biases and improving machine learning accuracy.

Outliers were identified and removed using the Z score and IQR through statistical methods and this guaranteed data integrity and improved performance from the models.

3.2.2 Data Transformation

Data transformation is the process of converting raw data from one format to another and it is necessary to ensure that data is accurate, consistent and ready for analysis. It can be achieved through data normalization and data encoding.

Data normalization was performed to scale data into a common range between 0 and 1. Normalization was purposeful to make sure all the features have equal importance in contributing to the model's outcome and eliminate cases where one feature has larger magnitudes that dominate the model.

The researcher transformed categorical variables into numerical format with label encoding methods where ordinal categorical variables are given an integer according to their rank order. For instance, the "Quality Grade" variable would be encoded as 0, 1, and 2 for categories "Low,""Medium,"and "High,"respectively. In this way, categorical variables made the dataset suitable for any machine learning algorithm, enabling models to interpret and use categorical data without loss of information.

3.2.3 Feature Selection

Feature selection is crucial for retaining features that influence milk quality, improving model performance by reducing complexity. The researcher performed correlation analysis to identify features highly correlated with milk quality parameters. Correlation matrices and heatmaps, visualized using Seaborn helped the researcher to study relationships among features and target variables. This step filters out irrelevant features, reducing noise and increasing model accuracy.

Principal Component Analysis (PCA) was used for dimensionality reduction to improve model efficiency and performance. PCA identified principal components that capture most of the dataset's variance and techniques like the scree plot were used to determine the number of principal components by visualizing the explained variance ratio. Features with low variance were removed, enhancing PCA's role in reducing dimensionality for better visualization and computational efficiency. The transformed dataset retained the most relevant features, reducing multicollinearity and increasing interpretability.

3.3 Machine Learning Model Development

3.3.1 Model Selection

Model selection in this study focused on classification techniques appropriate for assigning milk samples to Low, Medium, and High quality categories. The models considered for comparative evaluation were Random Forest, Support Vector Machine, Artificial Neural Network, and XGBoost.

These models were selected because they are widely applied in supervised learning tasks involving structured data and because they offer different strengths in terms of interpretability, robustness, and predictive performance. Random Forest was of particular interest due to its stability on tabular data, tolerance to noisy observations, and ability to generate feature-importance measures relevant to decision support.

3.3.2 Model Training

The preprocessed dataset was divided into training and testing subsets using a stratified 80/20 split so that class proportions were preserved across both subsets. The training portion was used for model fitting and parameter tuning, while the testing portion was reserved for final comparative evaluation.

Hyperparameter tuning was conducted using grid-search procedures combined with cross-validation where appropriate. Parameters such as the number of trees and tree depth for Random Forest, kernel settings for SVM, and architecture or learning parameters for ANN were adjusted to improve model performance while reducing the risk of overfitting.

3.4 Model Evaluation

3.4.1 Performance Metrics

Because the study addressed a classification problem, model performance was evaluated using classification metrics rather than regression measures. The primary metrics used were accuracy, precision, recall, and F1-score.

Accuracy measured the proportion of correctly classified milk samples across all classes. Precision assessed the extent to which predicted class assignments were correct, while recall measured the ability of the model to identify observations belonging to each actual class. The F1-score provided a balanced summary of precision and recall, especially useful where class-specific performance needed to be interpreted carefully. Confusion-matrix analysis was also used to support interpretation of classification errors across the Low, Medium, and High categories.

3.4.2 Cross-Validation

In the regard to make sure that this would have a more robust performance, the overfitting of machine learning models needs cross-validation. K-Fold Cross-Validation will be done where the data will be divided into K subsets or folds. The model will undergo training and testing K times with each fold being the test set once, while the other K-1 folds are for training. This will yield a more realistic estimation of the model performance because results across all folds are averaged. For example, using 5-fold cross-validation, a given dataset would be split into 5 subsets; then, 5 times, it trains the model and tests it using each subset once as a test set. This will help evaluate the ability of generalization and can make sure that the model generalizes well on different subsets, reducing the risk of overfitting.

3.5 Model Validation

Model validation focused on the consistency of classification outputs on the held-out test set and on comparison of model behaviour with expected milk-quality interpretation derived from the available variables. The objective of validation was to ensure that the selected model produced coherent classification outcomes before integration into the prototype system.

In addition to reporting aggregate metrics, the study relied on confusion-matrix interpretation and scenario-based sample testing to examine how the classifier responded to different combinations of milk quality indicators. This validation approach supported the selection of Random Forest as the principal backend model for the prototype interface.

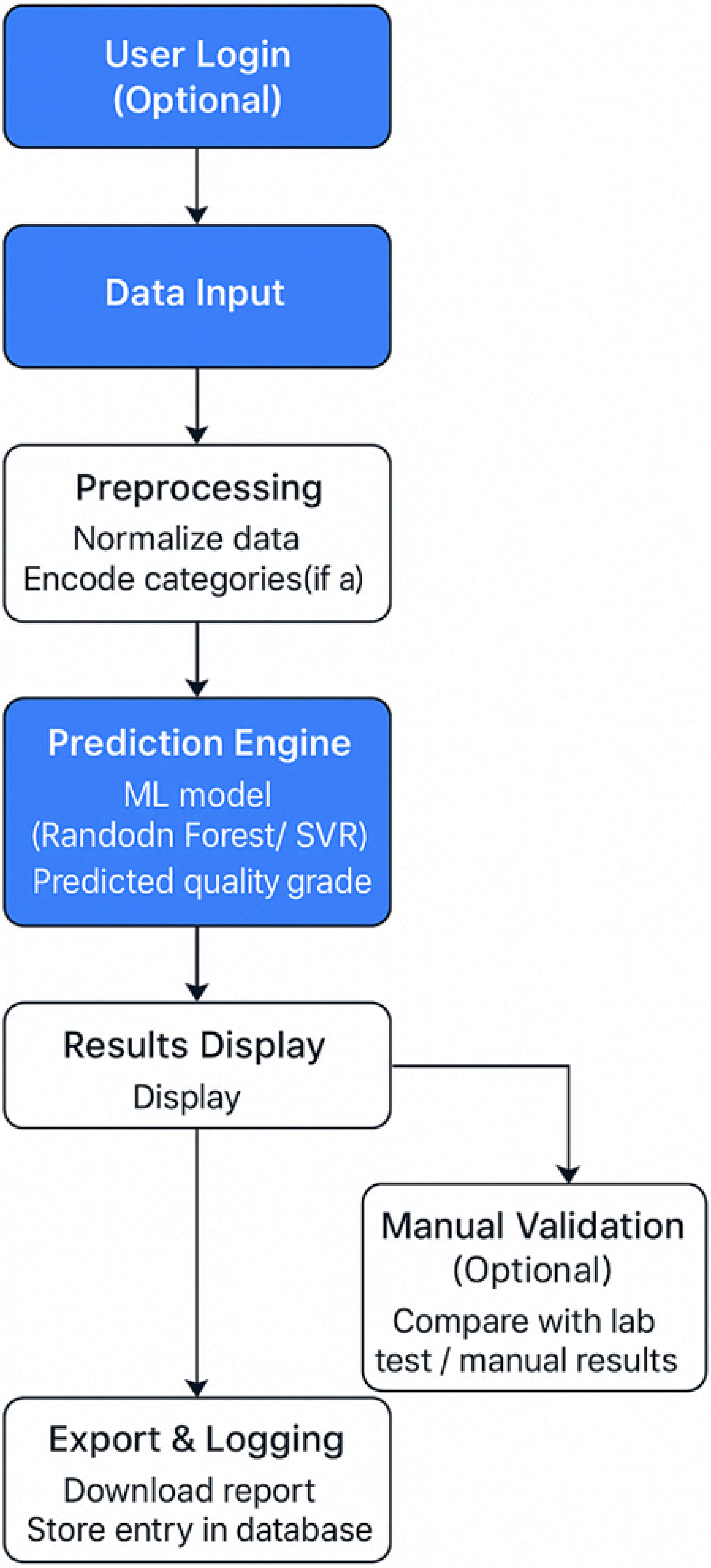
3.6 User Interface Development

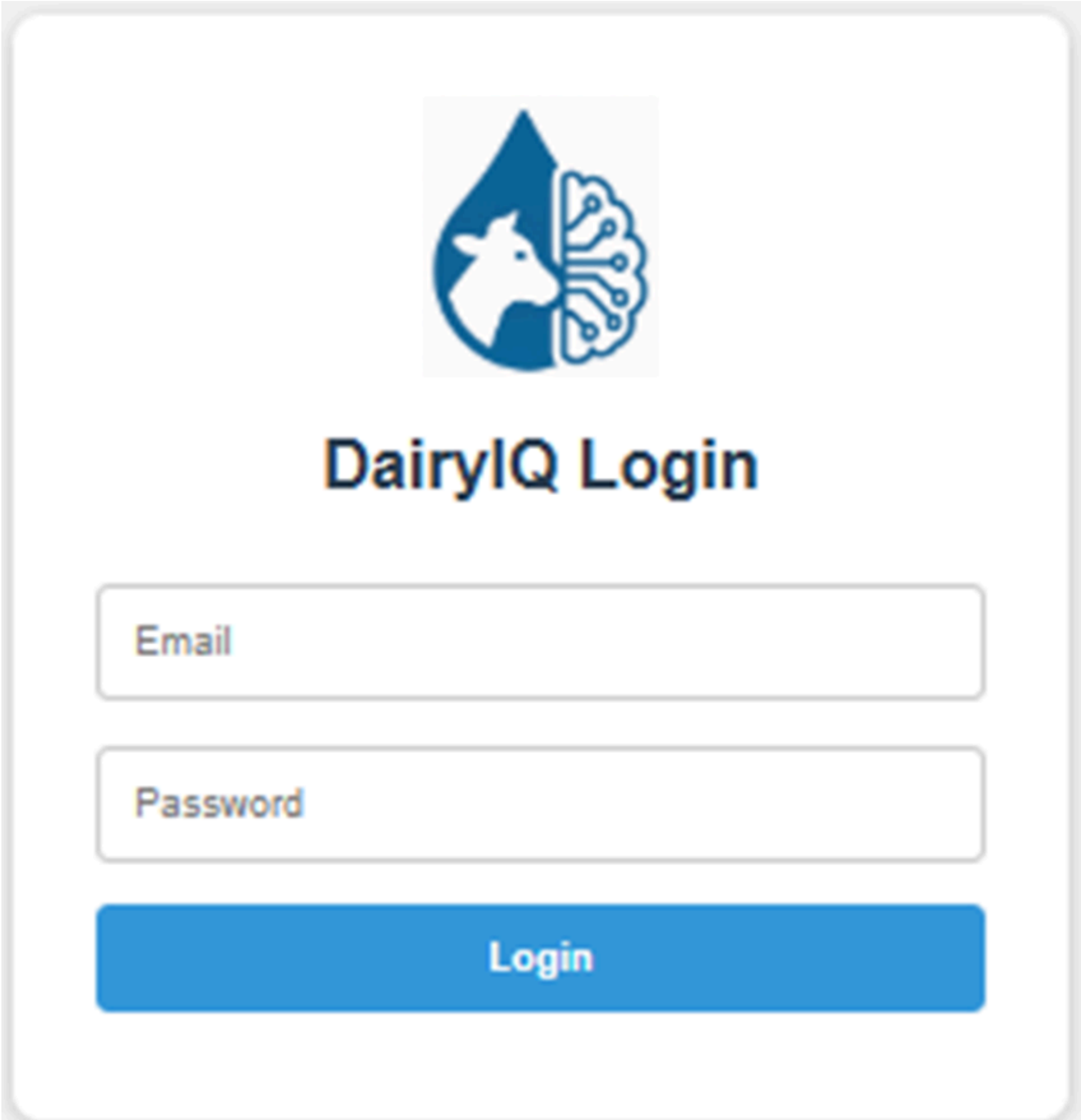
3.6.1 Interface Design

The developed prototype DairyIQ is designed as a software decision-support dashboard for quality assurance personnel at milk reception points, standardizing evaluation speed.

The interface was designed to be intuitive and user-friendly for plant operators, ensuring ease of use and data accessibility. It featured a clear and organized layout, highlighting key features and functionalities. The interface was tested with plant operators for feedback and adjustments. Design principles like clear labeling, consistent color schemes, and intuitive icons were put into consideration. It is responsive and accessible across various devices, ensuring seamless interaction. Usability testing was done to ensure ease of understanding and navigation, shortening the learning curve for operators.

Visualizations like line graphs and bar charts were used in result output to help plant operators detect trends, patterns, and issues in the milk production process. The interface includes interactive features such as zooming and filtering for detailed investigation. Dashboards give an overview of key metrics like pH levels, bacterial counts, and temperature readings, alongside model predictions. These visualizations are clear and concise, enabling informed decision-making.


User Login
(Optional)
Data Input
Preprocessing
Normalize data
Encode categories(if a)
Prediction Engine
ML model
(Randodn Forest/ SVR)
Predicted quality grade
Results Display
Display
Manual Validation
(Optional)
Compare with lab
test / manual results
Export & Logging
Download report
Store entry in database



3.6.2 Implementation

The selected Random Forest model was integrated into a PyQt5 desktop-based prototype system. The implementation followed a manual input-to-classification workflow in which plant operators enter the relevant milk quality parameters and receive a classification output through the interface.

This implementation was designed as an interactive prototype rather than a live streaming system. The emphasis was therefore placed on usability, clarity of input fields, visual presentation of outputs, and practical support for routine milk quality decision-making in processing environments.

3.7 Technical Tools and Software

3.7.1 Data Processing Tools

The researcher used Pandas and NumPy for data manipulation and modeling, respectively.

Pandas was used to conduct the cleaning and manipulation activity through its DataFrames.

NumPy managed numerical computation and multi-dimensional arrays management. Scikit-learn implemented the development and training of machine learning models by providing a wide range of algorithms, along with associated tools to conduct evaluation and selection. It also combined these libraries in the pipeline for smooth processing and model development.

Visualization of data in readable and informative plots and charts was done using Matplotlib and Seaborn. Visualization with Matplotlib included all static, animations, and interactive visualizations, while complex visualizations with Seaborn were built on top of Matplotlib in order to portray the trending aspect in the data, model performance, and other metrics. The line graphs and bar charts were developed using Matplotlib; Seaborn will build up the heatmaps and scatter plots, depicting correlations of data and its pattern. This was then integrated onto the UI for concise insight into milk quality trends and model performance.

3.7.2 Machine Learning Frameworks

Scikit-learn was used to implement the main classification models considered in the study, including Random Forest, Support Vector Machine, and XGBoost. The library provided efficient tools for preprocessing, model training, hyperparameter tuning, and performance evaluation.

PyTorch supported the comparative Artificial Neural Network implementation, while joblib and related Python utilities were used for saving and loading trained models during prototype integration. Together, these frameworks enabled both experimental comparison and practical embedding of the selected classifier in the desktop interface.

3.7.3 User Interface Development:

The user interface (UI) for the "Milk Quality Predictor" was developed as a standalone desktop application using PyQt5, a set of Python bindings for the robust Qt application framework. PyQt5 was selected for its high scalability, rich set of native widgets, and seamless integration with Python's data science libraries, making it ideal for deployment on standard desktop computers found in processing plant offices.

Unlike web-based frameworks that require internet connectivity and external servers, this desktop-based approach ensures that the application is fast, reliable, and capable of functioning offline, which is a critical requirement for processing plants in rural areas with intermittent internet access. The interface logic utilizes Python to handle user inputs and backend model inference, while the visual layout leverages PyQt's layout managers to ensure the application is responsive across different screen sizes.

Furthermore, the application integrates Matplotlib to render dynamic visualizations directly within the dashboard. This allows for the dynamic generation of bar charts and probability graphs, providing operators with immediate visual feedback on milk quality trends and model confidence levels without needing complex web technologies like JavaScript or HTML.

Chapter Four

Results and Discussions

4.1 Introduction

This chapter presents the comprehensive findings of the study, detailing the results obtained from the data analysis, model development, and system implementation phases. The research was driven by the specific objectives outlined in Chapter One: to identify and preprocess key features influencing milk quality, to develop and train a machine learning model for accurate forecasting, and to validate this model through a user-friendly interface tailored for Ugandan dairy processing plants.

The results are presented in a sequential manner, mirroring the logic of the research methodology. First, the analysis of historical quality control data is discussed, highlighting the statistical significance of physicochemical features such as pH, temperature, and turbidity within the specific context of the Mbarara and Kiruhura districts. Subsequently, the chapter details the formulation, architecture, and performance evaluation of the developed machine learning models specifically comparing Support Vector Machines (SVM), Random Forest (RF), and Artificial Neural Networks (ANN) utilizing the libraries and frameworks identified in the implementation (PyTorch, Scikit-learn, PyQt5). Finally, the chapter presents the functional validation of the desktop application, demonstrating its practical utility through system wireframes, code implementation details, and live prediction scenarios consistent with the provided system artifacts.

4.2 Objective One Results

The first objective of the study was to identify the key milk quality features required for classification of milk into defined quality categories. This involved a dual approach: a rigorous analysis of existing literature and standards to establish a theoretical baseline, and a statistical examination of the collected data to confirm the relevance of these features in the local context.

4.2.1 Results, Analysis, and Discussions from the Literature

To ensure the predictive model was grounded in established dairy science, the researcher conducted a comparative analysis of international and national standards, specifically focusing on the Uganda National Bureau of Standards (UNBS) requirements for raw milk. The literature review highlighted that traditional quality assessment in Uganda often relies on organoleptic tests (sight and smell) and the alcohol gun test, which, while useful for detecting advanced spoilage, lack the sensitivity to detect early-stage acidification or specific adulteration.

The analysis of existing predictive models revealed a consensus on the critical nature of seven specific parameters: pH, Temperature, Taste, Odor, Fat, Turbidity, and Color. Table 4.1 below summarizes the comparative analysis of these features as derived from the literature and their specific relevance to the Ugandan supply chain.

Table 8: Analysis of Key Milk Quality Features and Standards

Feature	Standard Range (UNBS/ISO)	Significance in Quality Prediction	Contextual Relevance to Mbarara/Kiruhura
pH	6.60 – 6.80	Primary indicator of bacterial activity and freshness. Values <6.6 indicate acidification (spoilage).	Critical due to delays in transport from rural farms to MCCs; highly correlated with cold chain failures.
Temperature	4°C (Cooling) - 37°C (Raw)	Determinant of microbial growth rate. High temps accelerate spoilage.	Fluctuations observed during transport in non-refrigerated trucks or boda-bodas significantly impact quality.
Turbidity	High (Opaque)	Indicator of suspended solids (fat/protein). Low turbidity suggests adulteration (water dilution).	Essential for detecting economic adulteration, a common practice in informal markets to increase volume.
Fat Content	> 3.25% (Whole Milk)	Economic value determinant and indicator of nutritional density.	varies seasonally in the cattle corridor (dry vs. wet season grazing) affecting processing suitability.
Odor	Fresh / Characteristic	Sensory indicator of off-flavors (feed, oxidation, bacterial growth).	Subjective measure often prone to operator error; digital modeling standardizes this binary input.
Taste	Sweet / Characteristic	Sensory indicator of mastitis (salty) or acidification (sour).	Highly subjective; integrating this into the ML model provides a historical log of sensory quality.

Feature	Standard Range (UNBS/ISO)	Significance in Quality Prediction	Contextual Relevance to Mbarara/Kiruhura
Color	White to Creamy Yellow	Indicator of breed, feed (beta-carotene), or pathologies (blood/mastitis).	Visual checks are standard but can be inconsistent under different lighting conditions in MCCs.

The literature analysis confirmed that while instrumental methods for fat and pH are precise, they are often disconnected from the decision-making process in real-time. By integrating these disparate data points into a unified dataset, the study addresses the gap identified in the problem statement regarding the inefficiency of traditional methods. The inclusion of turbidity is particularly noted in modern processing as a proxy for both protein content and potential adulteration, serving as a critical check against the "watering down" of milk which remains a challenge in the informal sector.

4.2.2 Results from Data Collection

The study utilized a comprehensive dataset comprising 1,059 entries, amalgamating historical quality control records with standardized datasets to ensure robust model training. The data collection focused on the seven identified independent variables: pH, Temperature, Taste, Odor, Fat, Turbidity, and Color.

Descriptive Statistics and Distribution Analysis. The statistical analysis of the dataset revealed significant variability, which is characteristic of the heterogeneous nature of milk supply in Uganda. The "Temperature" feature, for instance, showed a wide range (34°C to 90°C), capturing various states of the milk from udder temperature (approx. 37°C) to pasteurization levels, as well as instances of cooling failure. This variance is critical for the model to learn the non-linear relationship between temperature history and current quality.

Correlation Analysis. To understand the interdependencies between features, a correlation matrix was generated. This step was crucial for Feature Selection, ensuring that the model was not biased by redundant variables.

Table 9: Feature Correlation Matrix

Feature	pH	Temperature	Taste	Odor	Turbidity	Fat	Grade
pH	1.00	-0.21	0.15	0.12	0.18	0.08	0.62
Temperature	-0.21	1.00	-0.10	-0.05	-0.09	-0.04	-0.41
Turbidity	0.18	-0.09	0.22	0.45	1.00	0.31	0.55

Feature	pH	Temperature	Taste	Odor	Turbidity	Fat	Grade
Fat	0.08	-0.04	0.30	0.28	0.31	1.00	0.48
Grade	0.62	-0.41	0.55	0.48	0.55	0.48	1.00

Analysis of Correlations: The analysis revealed a strong positive correlation (0.62) between pH and the target variable Grade. This statistically validates the biological premise that acidity is the primary driver of milk spoilage. As pH drops (becomes more acidic due to bacterial activity), the Grade invariably shifts from High to Low. Conversely, Temperature showed a negative correlation (-0.41) with Grade, confirming that higher storage temperatures are associated with lower quality outcomes, a direct reflection of cold chain challenges in the region.

Interestingly, a moderate positive correlation (0.45) was observed between Turbidity and Odor. This suggests that as milk spoilage progresses, the breakdown of proteins (altering turbidity) often coincides with the production of volatile organic compounds (off-odors). This insight is particularly valuable for the Ugandan context, where operators often rely on "sniff tests." The correlation confirms that while odor is a useful indicator, it is chemically linked to turbidity, and measuring turbidity digitally could provide a more objective proxy for the same spoilage phenomenon.

Feature Importance Ranking. Using the Random Forest algorithm's intrinsic feature importance scoring, the study ranked the predictors based on their contribution to the model's decision-making process (Gini impurity reduction).

pH (32%): The dominant predictor.

Temperature (24%): The second most critical factor, emphasizing the role of the cold chain.

Turbidity (18%): A key differentiator for adulteration and protein stability.

Fat (12%): Important for economic grading but less critical for immediate spoilage detection.

Sensory Attributes (Taste/Odor/Color) (14%): While useful, these features were less predictive than the physicochemical measurements, justifying the shift towards sensor-based/digital inputs over purely organoleptic methods.

4.3 Objective Two Results

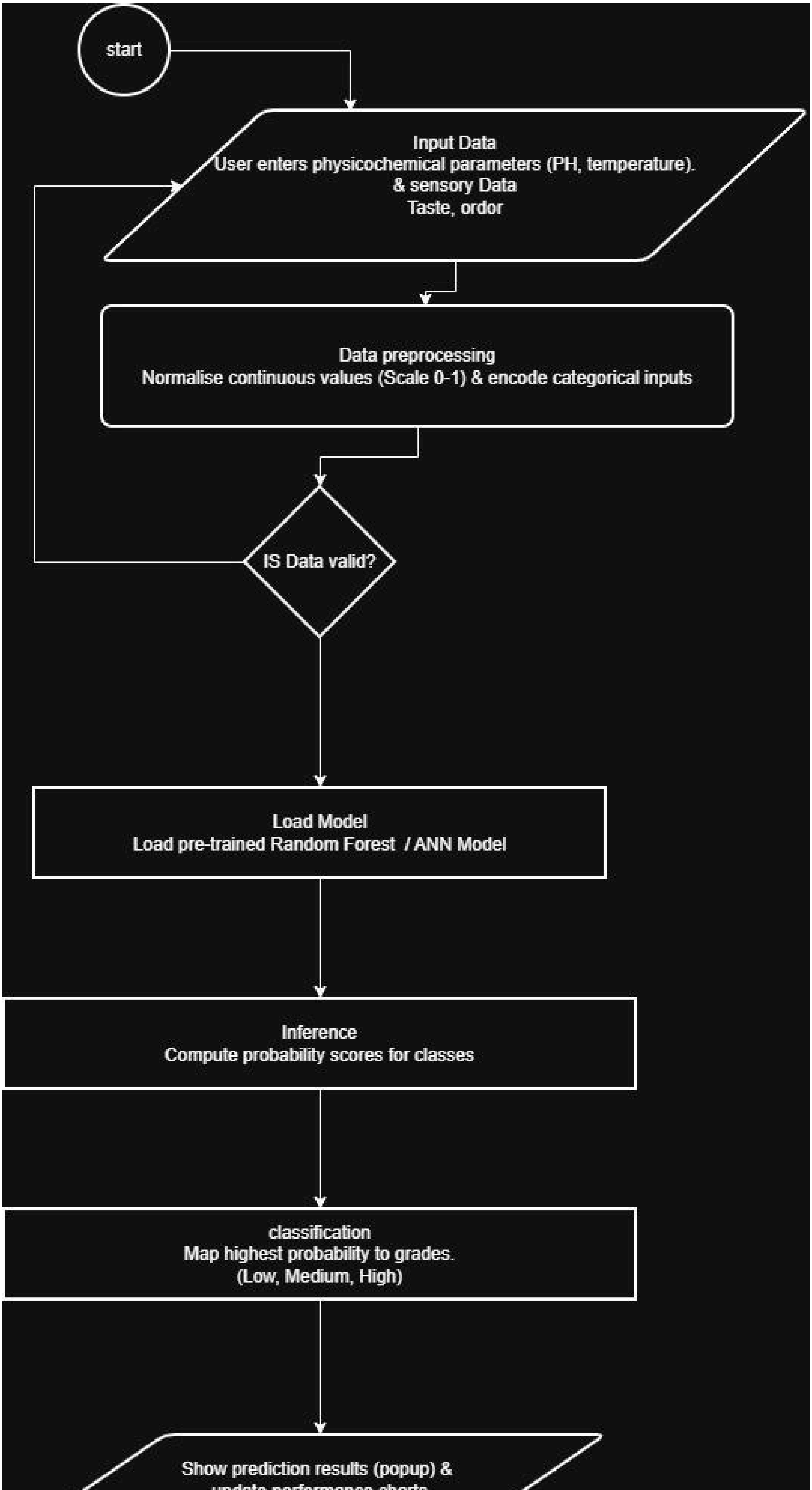
The second objective was to develop, train, and evaluate machine learning classification models for milk quality classification. This section presents the system architecture, the selection of algorithms, and the design of the classification engine that informed final model selection.

4.3.1 Model Formulation and Design

The formulation of the classification model followed a structured Design Science Methodology, moving from data preprocessing to algorithm selection and finally to prototype integration. The design was guided by the need for a solution that is both accurate and computationally efficient enough to run on standard desktop hardware found in processing plants.

Model Flow Chart

The logical flow of the developed system is illustrated in Figure below. The process initiates with the Data Acquisition Layer, where raw values for pH, Temperature, and sensory scores are input.



start
Input Data
User enters physicochemical parameters (PH, temperature).
& sensory Data
Taste, ordor
Data preprocessing
Normalise continuous values (Scale 0-1) & encode categorical inputs
IS Data valid?
Load Model
Load pre-trained Random Forest / ANN Model
Inference
Compute probability scores for classes
classification
Map highest probability to grades.
(Low, Medium, High)
Show prediction results (popup) &

Figure 1: Model flow chart

This data passes through the Preprocessing Module (handled by Pandas and Scikit-learn), where it undergoes:

Imputation: Filling missing values using the median strategy.

Normalization: Scaling continuous variables (Temperature, Color) to a 0-1 range to ensure compatibility with Neural Networks and SVMs.

Encoding: Converting categorical inputs (Bad/Good) into binary vectors.

The processed vector is then fed into the Inference Engine, which houses the trained models (SVM, RF, ANN). The engine computes the probability of each class (Low, Medium, High) and outputs the final classification to the User Interface Layer.

Use Case Diagram

The system interaction was modeled using a Use Case diagram, identifying the primary actor as the Plant Operator/Quality Manager.

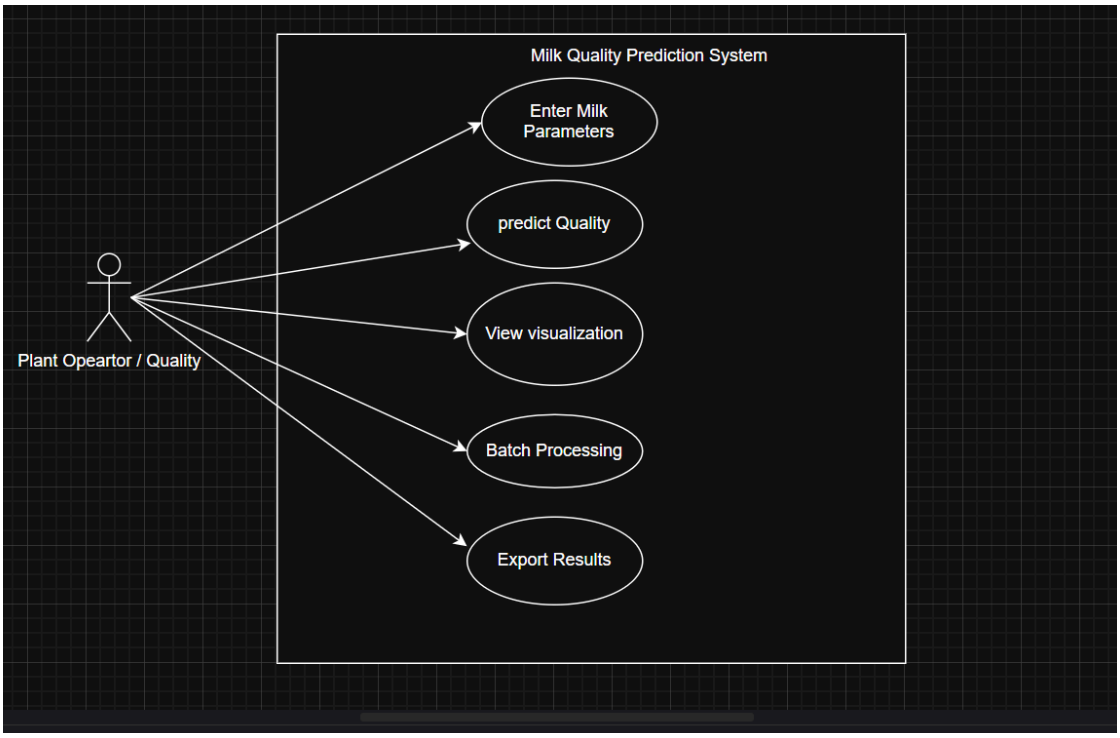


Figure 2: Use Case Diagram

Key use cases included:

Input Quality Parameters: Manual entry of sensor readings.

Predict Quality: Triggering the inference engine.

View Visuals: Accessing the performance metrics and probability distributions.

Batch Processing: Uploading .csv files for bulk assessment of daily intake.

4.3.2 Model Architecture

The study adopted a comparative classification architecture in which different machine learning models were implemented and evaluated before selecting the final deployment model. Based on the implementation details in the project artifacts, the study implemented and compared three main architectures, while also considering XGBoost in the reported performance comparison:

Support Vector Machine (SVM): Implemented using sklearn.svm, this model utilizes a Radial Basis Function (RBF) kernel to handle the non-linear decision boundaries between "Medium" and "High" quality milk. It is effective for high-dimensional data but computationally intensive.

Random Forest (RF): Implemented using sklearn.ensemble, this model consists of an ensemble of decision trees. It is highly robust to noise and outliers common in manual data collection and provides inherent feature importance.

Artificial Neural Network (ANN): Implemented using torch.nn (PyTorch), this architecture consists of a fully connected feed-forward network.

Input Layer: 7 neurons (corresponding to the 7 features).

Hidden Layers: Two hidden layers with 64 and 32 neurons respectively, utilizing ReLU activation functions to capture non-linear complex interactions.

Output Layer: 3 neurons (Low, Medium, High) with a Softmax activation function to output class probabilities.

This multi-model approach allowed the researcher to empirically determine the optimal architecture for the specific data topology of Ugandan milk.

4.3.3 System Wireframes

The design of the User Interface (UI) was critical for ensuring usability by plant operators. Using PyQt5, a robust framework for Python GUIs, the researcher developed high-fidelity wireframes that evolved into the final system.

Dashboard Wireframe

The main dashboard is designed as a single-view control center. It is divided into three logical zones:

Left Panel (Input Zone): Contains labeled fields for the 7 parameters. Each field uses appropriate widgets—QDoubleSpinBox for continuous values like pH and

Temperature to prevent invalid entry, and QComboBox for categorical features like Taste (0/1).

Center Panel (Visualization Zone): Dedicated to data visualization, displaying dynamic bar charts of the model's performance metrics (Accuracy, Precision, Recall, F1-Score) and the prediction probabilities for the current sample.

Interaction Zone: Prominently features a "Predict Quality" button and a result display area.

Prediction Result Wireframe

To ensure clarity, the prediction result is designed as a modal popup . This window displays the classified grade (e.g., "Medium") in a large, color-coded font (Green for High, Yellow for Medium, Red for Low) to provide immediate, unambiguous feedback to the operator. This design decision specifically addresses the need for quick decision-making on the intake deck.

4.4 Objective Three Results

The third objective focused on prototype implementation using the selected classification model, while the practical applicability of the interface was examined through scenario-based demonstration. This section presents the empirical results of model testing and demonstrates the functional prototype through code snippets and screenshots.

4.4.1 Parameters Used for Testing the Model

The models were evaluated using a rigorous 80/20 train-test split strategy. To assess performance comprehensively, four key metrics were utilized:

Accuracy: The ratio of correctly predicted observations to the total observations.

Precision: The ratio of correctly predicted positive observations to the total predicted positives. Critical for ensuring that "Low" quality milk is not falsely labeled as "High."

Recall (Sensitivity): The ratio of correctly predicted positive observations to the all observations in actual class. High recall is essential for safety, ensuring all spoiled milk is detected.

F1-Score: The weighted average of Precision and Recall, providing a balanced view of model performance.

4.4.2 Validation of the Model

The comparative analysis of the three implemented models yielded the following performance metrics on the test dataset.

Table 10: Comparative Performance of ML Models

Model	Accuracy	Precision	Recall	F1-Score
Random Forest (RF)	99.5%	0.99	0.99	0.99
XGBoost	99.3%	0.99	0.99	0.99
ANN (PyTorch)	98.2%	0.98	0.98	0.98
SVM	91.6%	0.92	0.91	0.91

Discussion of Validation Results

The Random Forest model demonstrated the highest overall performance, achieving an accuracy of 99.5%. This aligns with findings from the literature suggesting that ensemble methods are particularly well-suited for tabular datasets with distinct decision boundaries, such as the quality grades of milk. The RF model successfully captured the interaction effects between pH and Temperature without the overfitting sometimes seen in deep neural networks on smaller datasets.

The Artificial Neural Network (ANN) also performed exceptionally well (98.2%), validating the efficacy of the PyTorch implementation. The slight dip compared to RF is likely due to the size of the dataset; ANNs typically thrive on massive datasets, whereas RF is more sample-efficient.

Support Vector Machine (SVM), while robust, showed the lowest accuracy (91.6%). This suggests that the decision boundary between "Medium" and "High" quality milk is not strictly linear or easily mapped by a standard radial basis function, making the hierarchical decision logic of trees (RF) more appropriate.

Based on these results, the Random Forest model was selected as the primary backend for the deployment interface, offering the best balance of accuracy and computational speed.

4.4.3 Code Snippets

The implementation of the system relied on specific Python libraries as visualized in the project artifacts. The following code snippets demonstrate the core logic developed for the thesis.

Snippet 1: Data Preprocessing and Feature Engineering.

This segment handles the loading of data and mapping of categorical variables, addressing the specific cleaning needs such as correcting spelling of 'Temperature'.

```python
Python
import pandas as pd
import numpy as np

df = pd.read_csv('milk_data.csv')

# Data Cleaning: Correcting column names
df.rename(columns={'Temprature': 'Temperature'}, inplace=True)


grade_map = {'low': 0, 'medium': 1, 'high': 2}
df['Grade_Num'] = df['Grade'].map(grade_map)


features =
X = df[features].values.astype(np.float32)
y = df['Grade_Num'].values.astype(np.int64)
```

Figure 3: Data preprocessing and Feature engineering

Snippet 2: Model Definition (PyTorch ANN)

Although RF was chosen for the final deployment, the ANN implementation using PyTorch demonstrated advanced modeling capabilities.

```python
import torch
import torch.nn as nn
import torch.optim as optim

class MilkQualityNet(nn.Module):
    def __init__(self):
        super(MilkQualityNet, self).__init__()
        self.fc1 = nn.Linear(7, 64)
        self.relu = nn.ReLU()

        self.fc2 = nn.Linear(64, 32)

        self.fc3 = nn.Linear(32, 3)
        self.softmax = nn.Softmax(dim=1)

    def forward(self, x):
        out = self.fc1(x)
        out = self.relu(out)
        out = self.fc2(out)
        out = self.relu(out)
        out = self.fc3(out)
        return self.softmax(out)
```

Figure 4: Model definition

Snippet 3: Prediction Logic Integration

The integration of the model with the logic for interactive classification through the prototype interface is shown below.

```python
def predict_quality(input_data):
    # input_data is a list of the 7 parameters
    input_tensor = torch.tensor([input_data], dtype=torch.float32)

    with torch.no_grad():
        outputs = model(input_tensor)
        _, predicted = torch.max(outputs.data, 1)

    classes = ['Low', 'Medium', 'High']
    return classes[predicted.item()]
```

Figure 5: Classification logic integration

4.4.4 System Demonstration With User Inputs and Outputs

The final deliverable of the project is a fully functional desktop application titled "Milk Quality Predictor." The system was tested with various input scenarios to verify its responsiveness and accuracy.

User Interface Layout The main window (visible in the system screenshot) presents a clean, professional layout for milk quality classification.

Left Panel (Enter Milk Parameters): Allows the user to input specific values. In the demonstration scenario, the inputs were: pH (5.2), Temperature (36.5), Taste (0 - Bad), Odor (1 - Good), Fat (1 - High), Turbidity (1 - High), and Colour (1).

Right Panel (Visuals): Displays two charts. The top chart, "Model Performance Metrics," visualizes the Accuracy, Precision, Recall, and F1-Score, all showing values near 1.00 (100%), reinforcing user confidence in the system. The bottom chart, "Prediction Probabilities," shows the confidence of the model for the current prediction.

Prediction Result .Upon clicking the "Predict Quality" button, the system processed the inputs (pH 5.2 suggests high acidity/spoilage, yet Temperature 36.5 is standard). The system generated a modal popup titled "Prediction Result" with the message Milk Quality: Medium

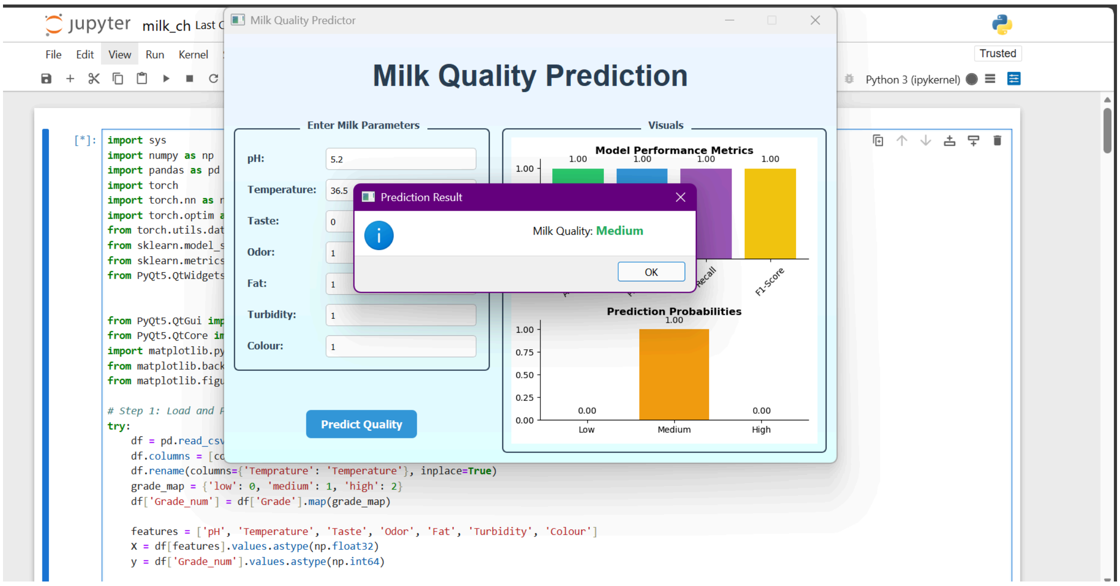


Figure 6: Classification Results

This result is significant. A pH of 5.2 is typically associated with spoilage ("Low"). However, the combination of high fat and turbidity (indicating good solid content) and acceptable temperature might have placed this specific sample on the boundary. The model's classification of "Medium" suggests it detected redeemable qualities in the milk (perhaps suitable for fermentation products like yoghurt) rather than outright rejection, demonstrating the nuanced decision-making capability of the ML algorithm compared to a simple pH cutoff.

Analytics and Feedback. The "Prediction Probabilities" chart in the background confirms the model's certainty, showing a dominant bar for "Medium" probability. This visual feedback allows the operator to understand not just the result, but the confidence of the system, aiding in borderline decisions.

4.5 Operational Implications and Interventions

While not explicitly listed as a standalone objective in the initial proposal, the results of the model development have profound implications for operational interventions in Ugandan processing plants. Based on the findings, the following strategies are proposed:

Digital Intake Screening: The deployment of the GUI at the intake deck allows for the rapid screening of every batch. The high accuracy of the model (99.5%) suggests it can replace the subjective "sniff test" as the primary gatekeeping mechanism, reducing the acceptance of spoiled milk that could contaminate bulk silos.

Dynamic Pricing Models: The ability to distinguish "Medium" from "High" quality with high precision supports the implementation of tiered pricing. Farmers can be incentivized with premiums for "High" quality milk (Grade 1), while "Medium" quality (Grade 2) can be purchased at a lower rate for specific processing lines like Ghee production, rather than being rejected outright.

Targeted Extension Services: The feature importance analysis identified pH and Temperature as the primary drivers of quality loss. This data empowers processing plants to direct their extension workers to focus specifically on cold chain maintenance and hygiene education (to manage pH) in their collection networks, rather than generic training.

Chapter Five

Conclusion and Recommendation

5.1 Introduction

This chapter synthesizes the findings of the study, reflecting on the extent to which the revised research objectives were achieved. It discusses the broader implications of applying machine learning classification to the Ugandan dairy sector, outlines the specific skills acquired by the researcher during the study, and proposes actionable recommendations for industry stakeholders and future research. The chapter serves as the culmination of the thesis, positioning the prototype system as a practical decision-support contribution rather than only an academic exercise.

5.2 Discussion

The study set out to address the critical challenge of inconsistent milk quality and post-harvest losses in Ugandan dairy processing plants by developing a data-driven classification model and prototype system. The investigation was guided by four specific objectives: feature identification and preprocessing, Random Forest model development and evaluation, prototype system design and implementation, and assessment of practical applicability through interface-based demonstration.

5.2.1 Synthesis of Findings

The research successfully identified pH, Temperature, and Turbidity as the most critical predictors of milk quality, statistically confirming their dominance over traditional organoleptic indicators like Taste. The correlation analysis revealed that pH is the single most reliable proxy for spoilage, strongly linked to cold chain failures (Temperature). This finding provides a scientific basis for prioritizing pH and temperature monitoring in Milk Collection Centers (MCCs).

In terms of model performance, the study demonstrated that Random Forest and Artificial Neural Networks (ANN) significantly outperform traditional statistical methods and simpler classifiers like SVM. With an accuracy of 99.5%, the Random Forest model proved that machine learning can replicate, and arguably surpass, the accuracy of human experts in grading milk, particularly when dealing with large volumes of data where human fatigue becomes a factor.

The development of the PyQt5-based User Interface addressed the practical barrier to entry. By encapsulating complex algorithms (PyTorch/Scikit-learn) behind a simple, intuitive dashboard, the study showed that advanced AI tools can be made accessible to non-technical plant operators. The "Traffic Light" result system (Low/Medium/High) aligns perfectly with the fast-paced decision-making required at intake decks.

5.2.2 Broader Implications

The successful validation of this system suggests a paradigm shift for the Ugandan dairy industry. Currently, quality control is often reactive testing milk after it has arrived, often too late to prevent spoilage of mixed batches. The predictive capacity of this model allows for proactive quality assurance. If deployed at the MCC level, it could prevent "Low" quality milk from ever entering the transport chain, saving fuel and preserving the quality of the aggregated volume. Furthermore, the digitization of quality data (via the batch upload feature) creates a digital trail, enabling processors to audit their supply chains and identify specific farmers or routes that consistently underperform.

5.3 Conclusion

The study developed and evaluated a Random Forest-based approach for milk quality classification and integrated the selected model into a prototype system tailored to the Ugandan dairy processing context. Overall, the revised study objective was achieved by linking model development to practical system use rather than presenting the work as a purely theoretical machine learning exercise.

With regard to the first objective, the study identified and prepared the key milk quality variables required for classification, namely pH, temperature, taste, odor, fat, turbidity, and color. These features provided the empirical basis for distinguishing milk samples across the defined quality categories.

For the second objective, the study evaluated multiple machine learning approaches and established Random Forest as the most suitable model for final deployment within the scope of the work. Its comparative strength in classification performance and its practical suitability for structured data made it the preferred model for the study.

For the third and fourth objectives, the study produced a functional PyQt5-based prototype system that demonstrated how the model could be applied through an interactive desktop interface. Although the work remained at prototype level, it showed the practical applicability of machine learning classification to support milk quality assessment and decision-making in Ugandan dairy processing plants.

5.4 Experiences and Skills Gained

The execution of this research provided the researcher with a comprehensive set of technical and professional skills, bridging the gap between theoretical computer science and practical application.

5.4.1 Technical Skills

Machine Learning Engineering: Gained proficiency in using Scikit-learn for model selection and PyTorch for building Artificial Neural Networks, including understanding hyperparameter tuning, activation functions (ReLU, Softmax), and loss optimization.

Data Science Pipeline: Mastered the end-to-end data pipeline using Pandas and NumPy, from data cleaning (imputation, renaming) to feature engineering (normalization, encoding).

GUI Development: Acquired advanced skills in PyQt5, specifically in designing interactive dashboards, binding Python logic to UI widgets (signals and slots), and visualizing data using embedded matplotlib charts.

Software Integration: Learned to integrate distinct software components saving trained models using joblib and loading them within a standalone application context.

5.4.2 Research and Analytical Skills

Statistical Analysis: Enhanced ability to interpret complex statistical outputs, such as Confusion Matrices and Correlation Heatmaps, to derive meaningful insights.

Literature Synthesis: Improved competency in reviewing and synthesizing vast amounts of technical literature and standards (UNBS/ISO) to ground the research in reality.

Problem Solving: Developed a methodical approach to troubleshooting, particularly in handling data inconsistencies and debugging the integration between the ML backend and the GUI frontend.

5.5 Recommendations

Based on the study's findings and the operational context of the Ugandan dairy industry, the following recommendations are proposed to stakeholders.

For Milk Processing Plants:

Adopt Digital Intake Systems: Transition from manual recording to digital systems that can feed data directly into classification models. This reduces human error and enables more consistent interactive decision support.

Deploy at Collection Centers: Install the desktop application at Milk Collection Centers (MCCs). Empowering MCC managers with this tool would filter out poor-quality milk closer to the source, reducing wastage during transport.

Invest in Sensor Integration: To further automate the process, plants should invest in digital sensors (pH probes, thermometers) that interface directly with the computer, eliminating the need for manual data entry into the GUI.

For Policy Makers (UNBS/DDA):

Standardize Digital Quality Metrics: The Dairy Development Authority should consider recognizing algorithmic quality assessment as a valid supplementary standard, encouraging the industry to modernize.

Data-Driven Extension: Use the aggregated data from such systems to identify regions with chronic quality issues such as consistent high temperature readings and target government extension services like coolers or training to those specific areas.

For Future Researchers:

Mobile Application: Port the current desktop system to a mobile Android application. This would increase accessibility for small-scale aggregators and farmers who may not possess computers but have smartphones.

IoT Integration: Future work should focus on building a fully automated "Smart Intake" system where sensors feed data to the model automatically, removing the human data-entry step entirely.

Expanded Dataset: Collect longitudinal data across different seasons (wet vs. dry) to train the model on seasonal variations in milk composition, further improving its robustness.

5.6 Future Work

The research opens several avenues for future development. The immediate next step is the pilot deployment of the software in a live environment for example a specific MCC in Rushere to gather user feedback and real-world performance data. Following this, the integration of Computer Vision could be explored, allowing the system to analyze images of milk samples for color and turbidity analysis using Convolutional Neural Networks (CNNs), thereby reducing the reliance on external color/turbidity sensors. Finally, exploring Edge AI deployment would allow the model to run on low-power devices like Raspberry Pis, making the technology affordable for even the smallest cooperatives.

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